

THE COMPACT — Visual Asset Bible

THE COMPACT · engine RULES_VERSION 40

July 31, 2026

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THE COMPACT — VISUAL ASSET BIBLE

Everything in this game that needs artwork, from the smallest object to the full screen, with the engine's own facts attached to each one. Written 2026-07-30 against `RULES_VERSION` 40.

Read-only pass. No engine file was changed to produce this.

0. How to read this document

You are going to draw this game. This document is the **inventory**: every object, every state, every variant, and — for each one — whether the data to draw it already exists.

It is written for someone who has **not** read the spec. Where a rule matters to the picture, the rule is stated here.

Five things are attached to every item:

CANON NAME	The one word the engine, the docs and the agents all use. Spelled exactly.
WHAT IT IS	What it means in the fiction, in one or two sentences.
FIELD	The exact field on the published JSON frame that carries it. If there is none, it says so.
STATES	Every variant you must be able to draw differently.
AT A GLANCE	What a stranger must be able to read off it in three seconds, with the sound off.
DATA TODAY	Whether a real world actually produces this, measured.

THE ONE RULE ABOUT NAMES

`SPEC.md` §3 is called "*Vocabulary — one word per concept*" and it is described in the spec as **a rules surface, not a style guide**. No word may name two concepts anywhere — not in code, not in a label, not in a legend, not on a mock. The project's worst inherited bug survived a full build and three review passes because the engine and the player-facing text disagreed about one word.

So: use the names in this document exactly. If you need a word for something new, it has to be added to the canon and something has to be removed, because the vocabulary budget is at its ceiling.

A short list of names that are already spent and mean something specific:

- **HOLDING** is your body on the map. Your assets are **STORES**. Never the reverse.
- **HAND** is one unit of presence. Not "worker", not "unit", not "ship".
- **HULL** is a warship. A **FIT** is what is on it. A **FORMATION** is a cohort of identical hulls.
- **CLAIM** is sovereignty over one system. **ANCHOR** is the structure that holds it.
- **WORKS** is the production building. A destroyed one is a **RUIN**.

- **CHARGE** is what a claim owes per Reckoning. **LEVY** is what the constellation owes. **RENT** is what a claim-holder takes from other people's WORKS. Three different words, three different flows.
- **STANDING** is a set of public vectors. It is **never a score** and must never render as one.
- **PARLEY**, **MESSAGE**, **DOSSIER** and **DISPATCH** are four different kinds of communication.

THE VERDICT SCALE USED THROUGHOUT

Verdict	Meaning
LIVE	The field is populated in a real world and the quantity it exists for actually varies. Draw it.
THIN	Rows exist, but the load-bearing number is constant, or some states never occur. Draw it, and expect to see only some of your variants until the world produces the rest.
EMPTY	The field exists and is wired, but zero rows in every world measured. Design the canvas; it will fill later.
NO FIELD	Named in the design, nothing on the frame. Do not design against it yet.

★ THE NAMED SIGNATURES — THE WHOLE ART BRIEF ON ONE PAGE

A13 is the axiom that makes this document necessary: **"Every mechanic renders. No named pixel signature → not ready."** Twenty-two signatures have been named. This is the index.

Signature	The picture, in the engine's own words	\$	Data
THE LODGE	the node sized by what its ground yields	6.4	LIVE
THE PINCH	a lane narrowed at its waist , notched with the detour	6.5	LIVE
THE VERGE	one closed fence per bloc; bare ground drawn bare	6.6	LIVE
THE CLAIM TINT	a claim tints a system, with three words on it	7.1	THIN
THE WORKS MARK	a WORKS marks a system; the mark carries the crowding	4.1	THIN
THE RUIN	a dark mark where production used to be, with who fell and when	4.3	EMPTY
THE PRINT	a market prints a price on a place, and the gap to everywhere else	6.8	THIN
THE TRIBUTE LINE	dashed · solid · red · reversing, thickness $\propto$ what is owed	7.2	LIVE
THE VENTURE RING	a ring whose hollow arc is the part riding on someone's word	8.1	LIVE
THE EMPTY SOCKET	an unfilled role pulses — <i>that is what forming looks like</i>	8.1	LIVE (live frame)
THE COMPACT LINK	a link between two holdings , and the snap that scars both	8.2	LIVE
THE AUTHORITY LINE	a directed line, thickness $\propto$ the authority handed over	8.3	LIVE
THE CLEARANCE PIPS	one pip per compartment the grant opens — <i>"that one can look"</i>	8.3	LIVE
THE DOSSIER THREADS	dashed arcs that never fade — the map's only dashed element	8.3	LIVE
THE CONVOY LINE	a line that can be severed — the map's motion, without its manifest	2.3	EMPTY
THE RAID LINE	a red arc with a countdown on it	9.1	THIN
THE BATTLE LINE	two rows of bars across a gap that moves every tick	5.4	EMPTY
THE SAP	a notched band whose advance is the score	9.3	EMPTY
THE FRONT BAND	a cone of odds before landfall; a swath of damage after	10.1	LIVE / partial
THE COVER ARC	filled for the escrowed half, hollow for the elective	10.2	EMPTY
THE COVER CHAIN	snaps at the link that broke; every link inward greys	10.3	EMPTY
★ THE RECEIPT REEL	the traitor's own words beside the promise it broke	12.3	EMPTY

Two more named in the design with **no field**: *a siege closes a ring* (Phase 1, honestly labelled) and the **drift mark** (mandate vs. deed).

THE PRIMITIVE VOCABULARY — REUSE IT, DO NOT EXTEND IT

The engine has already spoken for a specific set of visual primitives, and each one means one thing:

```
link · tint · ring · arc · line · bar · gap · tether · chain · halo · pip · socket · mark · print
```

A promise is an arc, filled for what is secured and hollow for what is not — and that grammar is used in *two* places deliberately (the venture ring and the cover arc), because they are the same idea about different subjects. **Reuse it. Do not invent a third promise shape.**

1. What a renderer actually consumes

There is no socket, no streaming API, and no database. **The client is a pure static-file consumer.**

File	Cadence	Cache	What it is
<code>frames/live.json</code>	every tick	<code>max-age=2</code>	The motion between ceremonies. 17 keys.
<code>frames/latest.json</code>	once per Reckoning	<code>max-age=2</code>	The full settled record of the day. 29 keys.
<code>frames/r-000015.json</code>	written once, never changes	<code>immutable</code> , 1 year	The archive. Note the six-digit zero padding .
<code>frames/index.json</code>	once per Reckoning	<code>max-age=2</code>	<code>{ reckonings: [{ reckoning, tick, file, stateHash, levyShort, kept, broken, beats }] }</code>

Three consequences worth designing around:

1. **Same frame in, same picture out.** No client-side simulation, no drift, no interpolation of authoritative facts. Two viewers with the same file see the same world.
2. **A free, exact scrubber.** The whole history is on disk as immutable files and there is no pruning at all. Scrubbing back through every Reckoning the world has ever had costs nothing and is exact rather than reconstructed.
3. **`stateHash` is a free change detector.** Two polls with the same hash are the same world.

THE TWO CLOCKS

This is the single most important fact for motion design.

```
TICK          = 5 minutes in production (2s "turbo", 10s "fast" in test worlds)
RECKONING     = 288 ticks = exactly 24 hours in production
SEASON        = 4-8 weeks
```

So `live.json` changes 288 times a day and `latest.json` changes once. Anything that animates — countdowns, the battle gap, convoys in flight, a venture socket pulsing — lives on the live frame. Anything that is a *result* — the rundown, the docket, standings, the Levy's outcome — lives on the Reckoning frame.

Within a Reckoning the clock has four named phases, published as `live.phase`:

phase	When	What it means on screen
EARLY	ticks 1–263	The long stretch. Work happens.
COMMITMENT	last 24 ticks	Ventures may still be joined and amended; late commitments go dark.
FREEZE	the single tick before settlement	Nothing may touch the settlement set.
SETTLING	the settlement tick	The one tick a Reckoning frame is written on.

`live.ticksUntilReckoning` counts 288 → 1. It is the countdown a viewer reads.

A9 — THE RULE THAT CONSTRAINS EVERYTHING YOU CAN SHOW

The spectator client never shows a live fact an agent's own `observe` would not show.

This is a hard rule, not a preference, because the agents read the public feed: any viewer privilege is instantly an agent exploit. Concretely, **you may draw only what is on the frame**, and the frame is refused rather than published if it carries a field nobody argued for.

The practical shape of it, which will come up constantly:

A ship at sea is visible; its manifest is not.

A convoy on a lane is public — it is the map's motion and the map is the show. *What it is carrying* is not. Same for a battle: hull counts and damage **fractions** are public; absolute hit points are not, because dividing them would reveal the fit, and a fit is a manifest.

THE FIVE VISIBILITY TIERS

Every fact in the game sits on one rung of this ladder. It decides what may be drawn, and when.

Tier	Agents now	Viewers now	Later
PUBLIC	everyone	yes	—
PARTIES	the parties only	no	everyone at settlement
SENSED	whoever has a hand in range	no	after the Reckoning it mattered in
SEALED	nobody	the flag only	content in the season replay
PRIVATE	the principal itself	never	never

Measured on a real 900-tick world, the ledger held PUBLIC=654 · PRIVATE=14 · SENSED=4 · PARTIES=0 · SEALED=0 . So in practice almost everything a world produces today is public.

THE LEGIBILITY BUDGETS — THESE ARE ENFORCED, NOT SUGGESTED

A frame that breaks one of these is **refused and not published**. They are hard caps on how much you can be asked to draw at once.

Budget	Value	What it caps
MAX_LABELS_PER_FRAME	7	Named entities on screen at once. The honest human maximum.
MAX_DOCKET_CARDS	7	Cards in the default view
MAX_RUNDOWN_SEGMENTS	12	Beats in the nightly broadcast
SEGMENT_SECONDS	30–45 s	Per beat. Human time — never scaled by the tick.
MAX_AUTHORITY_LINES	12	Grant lines. <i>"Convergence is the signature; a hairball is not."</i>
MAX_RAID_LINES	6	<i>"A countdown a viewer can follow, not a weather map."</i>
MAX_FRAME_CLAIM_LINES	12	Claim tints
MAX_FRAME_WORKS_LINES	16	Works marks
MAX_FRAME_MARKET_LINES	16	Price prints
MAX_FRAME_SAP_LINES	6	Campaigns. A floor under live wars , not a truncation.
MAX_FRAME_BATTLE_LINES	4	The smallest budget of all: <i>"a battle gets the screen, and the point of the budget is not that many do."</i>
MAX_BATTLE_FORMATIONS	12	Bars inside one battle (6 per side)
MAX_FRAME_RUINS	8	Ruins. Overflow drops the oldest .
MAX_FRAME_SYNDICATE_LINES	8	Orgs
MAX_FRAME_CONVOY_LINES	16	Convoys. Overflow keeps the ones landing soonest .
MAX_FRAME_COMPACT_LINKS	12	Links. Overflow never drops a snap .
MAX_FRAME_SWAY_LINES	32	THE VERGE. A ceiling on the map, not a selection — a fence cannot be sampled.
MAX_FRAME_FRONT_BANDS	12	Storm cells
MAX_FRAME_COVER_ARCS	8	Cover arcs
MAX_FRAME_COVER_CHAINS	4	Reinsurance chains
ticker line	140 chars	Every ticker line, hard

The direction a cap drops in is itself a design decision and it is written into each one. Ruins drop the oldest. Convoys keep the imminent. Compact links never drop a broken one. This is because a cap that drops the newest thing hides exactly the event the field exists to show.

2. THE ATOMS — goods, currency, lots

2.1 THE GOODS

There are **four goods**, and they are the whole material economy. Their identifiers are lowercase strings on the frame.

Canon name	id	What it is	Where it comes from	What it pays for
ORE	<code>ore</code>	The raw good a WORKS pulls out of the ground, at every tier. Settles nothing.	<code>extract</code> at a WORKS	Nothing directly — it is the <i>input</i> to both recipes
RATION	<code>ration</code>	The good every obligation is priced in	<code>refine {kind:"RATION"}</code> from ore	The LEVY, the CHARGE, a WORKS build, MATERIEL, a hull's frame, the endowment
ALLOY	<code>alloy</code>	The manufactured good. Payable against no obligation at all.	<code>refine {kind:"ALLOY"}</code> from ore, at a rate set by tier: COMMONS 8:1 · MARCHES 32:1 · FRONTIER 64:1	An ANCHOR (500 units), and a crossing beyond the Commons
FUEL	<code>fuel</code>	The Frontier-only third good	Only from Frontier ground (9–11/tick per system)	What an ANCHOR burns per Reckoning to keep collecting RENT, and half of what a HULL is built from

A correction worth stating loudly, because it will otherwise cost you an icon: MATERIEL is not a fifth good. It is a canon *term* for what a campaign destroys at its depot per pulse, and it is denominated in `ration` (`MATERIEL_GOOD = 'ration'`, 3,000 per pulse). Draw it as rations being burned by a war, not as a distinct substance. The same is true of **CHARGE** and **LEVY** — both are obligations denominated in `ration`.

AT A GLANCE: four icons, and the pair `ore → ration` / `ore → alloy` should read as a *conversion*, because the whole tension of the production chain is that raw yield cannot pay a debt. The headline meter `meters.unrefined` (measured **66,130** on a real world) is exactly *"wealth that cannot pay a debt"* — it rises when the world is extracting hard and converting slowly, which is the moment before somebody defaults while visibly rich.

DATA TODAY: LIVE. All four goods move in real worlds. `ore` and `ration` are everywhere; `alloy` and `fuel` are rarer and Frontier-shaped.

2.2 CURRENCY

- All money is **Minor** — integer minor units. There are **no floats anywhere** in this codebase's value paths, deliberately, so an amount is always exact and always an integer.
- All proportions are **basis points** (**Bps**), integer, `10_000 = 100%`. Anything you see named `...Bps` is a percentage $\times 100$. `electiveBps: 3501` is 35.01%.
- The engine's own display convention (`render.ts:money`) rounds for legibility: `≥1,000,000 → "1.4M"`, `≥1,000 → "12K"`, below that the exact number.

One important exception, and it should be visible in the type treatment: the accusation sentence is never rounded. When a principal walks away from a promise, the withheld amount prints in full and only the context is abbreviated — *"vex walked away from 1,500 of the 4K it had promised."* The reason is written into the source: rounding 1,500 up to "2K" overstates a permanent public accusation by a third, in the direction that harms the named principal. **Exact numbers accuse; rounded numbers describe.**

2.3 THE LOT — WHERE GOODS PHYSICALLY ARE

Everything is located. A **LOT** is a located quantity of one good in one account.

Field	Meaning
<code>good</code> , <code>qty</code>	What and how much
<code>location</code>	Which system it is standing in
<code>state</code>	<code>AVAILABLE</code> or <code>IN_TRANSIT</code>
<code>carrier</code>	The HAND carrying it — non-null if and only if <code>IN_TRANSIT</code>
<code>encumbranceId</code>	Exclusive. A pledged lot cannot back a second obligation, and cannot be sent away.
<code>origin</code>	Provenance — who created it

The two states, and how they must differ visually:

	<code>AVAILABLE</code>	<code>IN_TRANSIT</code>
Where it is	standing at a system	on a lane, aboard a hand
Can it be spent	yes	no
Can it be raided	yes	no — an in-transit hand is never routed today
Who is carrying it	nobody	exactly one named HAND
How it renders	a stack at a place	it does not render as goods at all — it renders as the CONVOY LINE , and the quantity is deliberately withheld

That last row is the §11.2 split in one line. **An in-transit lot must never be drawn as a visible quantity**, because the whole intel market depends on a raider having to scout for a manifest. The convoy line publishes the carrier, the two endpoints, and the arrival tick — and nothing else.

An **encumbered** lot deserves its own treatment: it is goods that exist, are yours, and are locked behind a promise. A hatch, a chain, a desaturated stack.

DATA TODAY: LIVE for `AVAILABLE`. `IN_TRANSIT` is real in the engine (`haul.departed` = 4, `haul.landed` = 4 on a measured world, matching) but `convoyLines` was 0 in every archived frame measured — see §14.

3. HANDS — the unit of both work and force

3.1 WHAT A HAND IS

| A **HAND** is one unit of simultaneous physical presence.

Every principal starts with exactly **three** (`HANDS_PER_PRINCIPAL` = 3) and that is the number for almost everybody, almost always. A hand works a site, escorts a load, garrisons a holding, carries cargo, or fills a venture role. **It cannot do two at once, and travel takes real time.**

Three rules that decide how it should look:

1. **Hands are rows, not a count.** They have identities, because the feed needs to say *"Vale's second hand fell at Orison"*. Draw three distinguishable hands per principal, not a number badge.
2. **A hand is never destroyed.** A hand taken in a raid goes `RECOVERING` at its holding for 12–48 ticks. Loss is cargo, time and position — never capacity. So there is no death state to draw, only a **healing** state.
3. **A hand is the unit of FORCE too.** `FORCE_PER_HAND` = 1. When you see two sides of a raid or a campaign counted, they are being counted in hands. So the same object is your labourer, your trucker and your soldier — and that is the scarcity the whole game turns on.

3.2 THE STATES

`HandState` has exactly four members.

State	What it means	Where the hand is	Visual job
IDLE	Free. The only state that can be given a new job.	standing at <code>location</code>	the default, and it must read as <i>available</i>
IN_TRANSIT	On a lane. <code>location</code> stays the origin until arrival resolves; <code>destination</code> and <code>freeAtTick</code> are set.	between two systems	motion, with a countdown
COMMITTED	Filling a venture role. Commitment lives in exactly one place — <code>venture_role.filled_by_hand_id</code> — never on the hand.	at the venture's stage	tethered to a ring
RECOVERING	Routed in a raid. Returns to its holding for <code>HAND_RECOVERY_TICKS</code> (12–48).	at its holding	wounded, counting down, not dead

A fifth thing to draw that is not a state: a hand carrying cargo versus a hand empty. The engine calls this `laden`, and it is a **single bit** on the convoy projection. A laden hand on a lane draws a CONVOY LINE; an empty one draws **nothing at all**, because a bare `move` emits no public event and publishing it would show a fleet redeployment no agent can see.

AT A GLANCE: how many of my three hands are free right now. That one number is the constraint the entire game is played against, and it is what makes the Levy bite — paying it requires simultaneous presence you do not have.

DATA TODAY: LIVE. Hand state drives convoy lines, raid force, battle formations and tribute lines.

4. STRUCTURES — WORKS, ANCHOR, THE RUIN

Three built things stand on the map. They are the only permanent marks a principal can make on geography.

4.1 THE WORKS — THE PRODUCTION STRUCTURE

CANON: `WORKS`. Never “mine”, never “factory”, never “site” (a **SITE** is a resource node, which is a different thing).

WHAT IT IS: A building you raise on a system that extracts `ore` from the ground every tick. It costs **5,000 ration** plus **25,000** in currency to build and takes **24 ticks to spin up**.

FIELD: `worksLines[]` on the Reckoning frame.

Field	Real value from a measured frame	What it draws
<code>works</code>	<code>works:sys-05:15:p:thessaly</code>	identity
<code>system</code>	<code>sys-05</code>	where the mark sits
<code>holder</code>	<code>p:thessaly</code>	whose it is
<code>yieldPerTick</code>	<code>110</code>	what the PLACE yields — a property of the map (see THE LODE)
<code>occupants</code>	<code>3</code>	live WORKS standing there. The crowding, which is the economic story
<code>sharePerTick</code>	<code>36</code>	what this holder actually keeps per tick: the share less the rent
<code>extracted</code>	<code>27421</code>	cumulative units handed over, gross. Never a stock reading.
<code>rentBps</code> / <code>rentPerTick</code> / <code>rentPaid</code>	<code>0 / 0 / 0</code>	what the landlord takes
<code>fuelPerTick</code> / <code>fuelExtracted</code>	<code>0 / 0</code>	the Frontier's third good
<code>legend</code>	<code>"EXTRACTING"</code>	the two words a viewer reads

STATES: the legend is either `EXTRACTING` or `SPINNING UP N ticks`. There are only two, and the frame refuses to publish a WORKS that reads `EXTRACTING` while quoting a dead share, or the reverse.

AT A GLANCE: *how crowded is this ground.* `occupants: 4` on one system is visibly a contested seam, and the whole mechanic exists to produce that picture. The arithmetic a viewer can check is `sharePerTick + rentPerTick = yieldPerTick / occupants`, and the frame is refused if it does not hold.

What a WORKS mark may never carry: the holder's stockpile, units in store, or anything derived from what the holder still *has*. `extracted` says what a place has **given up**; that is the safe side of the line and the only side.

DATA TODAY: THIN. 16 rows in every real frame, but `EXTRACTING` on every one, and the three rent fields measured **zero on all 144 rows** across the audited worlds. Design the rent treatment; expect to see it dark for now.

4.2 THE ANCHOR — WHAT HOLDS A CLAIM

CANON: `ANCHOR`. Never “berth”, never “mooring”, never “fleet position”.

WHAT IT IS: The produced goods destroyed *at* a system to bring a **CLAIM** into being, and then the thing that burns FUEL to keep it collecting. `build {kind:"ANCHOR"}`.

THE PRICE, exactly:

To raise	5,000 <code>ration</code> (<code>ANCHOR_QTY</code>) + 500 <code>alloy</code> (<code>ALLOY_ANCHOR_QTY</code>)
Bond posted	50,000 currency, slashable
Fuel burned	1,200 <code>fuel</code> per Reckoning on the FRONTIER; 0 in the Marches

FIELD: An anchor has **no field of its own**. It renders through its claim — `claimLines[]` — and specifically through `claimLines[].anchorHot`, a boolean.

STATES — and `anchorHot` is the one to draw:

	<code>anchorHot: true</code>	<code>anchorHot: false</code>
Meaning	fuelled for this Reckoning	cold — it is collecting nothing
What still holds	everything: the claim, the arrears count, the bond	everything: the claim, the arrears count, the bond
What is lost	nothing	the income only

This is a genuinely good picture and worth designing well: *a frontier landlord's rent stops while its tenants keep their whole share*. A supply failure whose cause is visible on the same screen. It is `true` by construction where no fuel is required, so a Marches claim never draws as switched off.

DATA TODAY: THIN — `claimLines` is empty in the short worlds and `SUPPLIED` on 24 of 24 rows in the longest audited world, so `anchorHot: false` has not been observed.

4.3 ★ THE RUIN — A RAZED WORKS

CANON: `RUIN`.

WHAT IT IS: A destroyed WORKS that **stays on the map forever**, labelled with who built it, who ended it, and the Reckoning it fell.

This is the clearest piece of visual-design reasoning in the whole engine and it is worth quoting the argument, because it generalises:

The obvious rendering of a razed WORKS is that its mark *disappears*. That is not a signature — **a mark that vanished is indistinguishable from a mark that was never there**. The system that stopped producing would look exactly like the system that never produced.

So a razing leaves something behind. **A ruin never leaves**, and a system carrying both a WORKS and a RUIN is a place that was fought over and rebuilt — which is the one thing a map of current facts can never show.

FIELD: `ruins[]`, newest first, capped at 8, overflow drops the **oldest**.

Field	Meaning
<code>works</code> , <code>system</code>	which, where
<code>holder</code> + <code>handle</code>	who built it. A ruin is named for its builder
<code>razedBy</code> + <code>razedByHandle</code>	who ended it, or <code>null</code> when the world did
<code>fellAtReckoning</code> , <code>fellAtTick</code>	when
<code>extracted</code>	cumulative units the place handed it before it fell — <i>the epitaph, as a number</i>
<code>legend</code>	<code>RUIN · fell R12 to vex</code>

A hard rule on the label: when `razedBy` is `null` the raid was world-spawned and **has no author**. The legend prints *"to the world"*. A frame that names a handle for an unauthored razing is refused, because it would be a permanent public claim that a real agent did this.

AT A GLANCE: a dark mark where production used to be, with a date and a name on it. Two ruins beside one live WORKS should read as a history, not as a bug.

DATA TODAY: EMPTY. Zero ruins in every frame of every world measured, over 8 seeds × 9 Reckonings. The mechanism is built and tested; nothing in the current world selects it. Design the canvas.

5. HULLS AND COMBAT

Combat is the one thing in this game that is **optional all the way down**. Nothing on the road to anything else needs a hull, and nothing but a hull needs a hull.

That makes it the layer with the most art per unit of gameplay, and the owner has explicitly put combat depth in scope. This section is deliberately generous.

A battle never exists on its own. It is what a **refused demand** becomes: a raid is declared, the target refuses to pay, and the standoff becomes an **ENGAGEMENT**. `BattleLine.raid` names the standoff it is fought inside, always.

5.1 ★ THE FIVE HULLS

CANON: `HULL` — a warship: a located, owned, priceable asset built from goods and **destroyed permanently**. Never “ship class in the abstract”; never a structure; and the last tank layer is **STRUCTURE**, not “hull”.

There are exactly **five hulls, one per size band**. The enum key *is* the identifier — there is no separate “class” word — and `size: 1..5` is the ladder.

id	Size	Role in the fiction	Structure / Shield / Armor	Sig	Mob	Cap / regen	Hardpoints	Slots H·M·L·R	Cost
PIKE	1	"The cheap ship that decides which expensive one leaves." Frigate.	150 / 150 / 100	40	9	300 / 30	2	2·3·2·1	400 ration + 20 fuel
LANCE	2	"Anti-small, and the world's own hull. Four hardpoints on a fragile frame: it deletes a screen and dies to anything that catches it." Destroyer.	260 / 260 / 180	70	7	400 / 34	4	4·3·2·1	700 + 40
WARDEN	3	"The compositional backbone, and the only hull with the CPU for REMOTE_REPAIR plus a real tank." Cruiser; the default doctrine chassis.	600 / 600 / 400	130	5	900 / 60	3	3·4·4·2	1,500 + 90
BULWARK	4	"The command platform." Battlecruiser. The only hull with command: 1 — and its command channel stops when it dies.	1,200 / 1,100 / 900	230	3	1,400 / 80	5	5·4·5·2	3,000 + 200
CITADEL	5	"The heavy line. Enormous alpha it cannot apply without help." Battleship. Costs 16× a PIKE and cannot hit one .	2,400 / 2,200 / 1,800	400	2	2,200 / 110	6	6·5·6·2	6,400 + 450

Each hull carries **one role bonus**, which is a reason to want it: **PIKE** +50% **TACKLE** · **LANCE** +30% **WEAPON** · **WARDEN** +50% **REPAIR** · **BULWARK** +50% **EWAR** · **CITADEL** +50% **DAMAGE**.

AT A GLANCE, and this is the whole silhouette brief: size, and what it is for. A **PIKE** must read as small, fast and dangerous to expensive things. A **CITADEL** must read as enormous and helpless against small things. The bare-frame ladder is **400 / 700 / 1,600 / 3,200 / 6,400** — a clean 16× spread — and mobility runs the other way, **9 → 2**.

Five more facts a hull asset needs:

Build time	4 ticks (HULL_FIT_TICKS)
Crew	exactly one IDLE hand per hull, at the stage. So a principal owns up to 6 and can fly 3
Tier	never in the COMMONS (A8), and only where its owner's holding stands
Mobility on the map	a hull never travels. It is berthed where it was built and fights only at that system
Refit	there is none, ever. The FIT is frozen at build and identified by a content hash

Hull states: **FITTING** → **READY** → **ENGAGED** → **WRECKED** . **WRECKED** is terminal and **the row is never deleted**.

The starter hull the game offers a new agent is a **PIKE** on the fit **SMALL_GUN** · **POINT** · **WEB** · **AFTERBURNER** .

5.2 MODULES AND FITTING

CANON: **FIT** — the immutable set of **MODULES** on a **HULL**, identified by a content hash, frozen at build. **MODULE** — one fitted item, consuming CPU and powergrid in one of four slot rows. **RIG** — a module in the fourth row, consuming calibration and **destroyed if removed**.

There are 32 modules in four rows. (Note: two doc comments in the engine say “twenty-nine”. The code has 32. Do not build a menu graphic off the prose.)

The four rows — **SLOT_ROWS** = ['HIGH', 'MID', 'LOW', 'RIG'] . **The twelve families** — **WEAPON** · **TANK_BUFFER** · **TANK_ACTIVE** · **TACKLE** · **EWAR** · **REPAIR** · **CAPACITOR** · **PROPULSION** · **COMMAND** · **FITTING** · **DAMAGE** · **APPLICATION** .

HIGH row — 9 modules. Guns, repair, drain, command.

id	Family	cpu / grid / cap	What it does
SMALL_GUN	WEAPON	8 / 6 / 4	hardpoint; kinetic
SMALL_GUN_EM	WEAPON	9 / 7 / 5	hardpoint; EM
LARGE_GUN	WEAPON	14 / 90 / 12	hardpoint; explosive
LARGE_GUN_EM	WEAPON	16 / 96 / 14	hardpoint; EM
MISSILE	WEAPON	20 / 18 / 0	hardpoint. Cap-free — which is why DRAIN cannot answer missiles
MISSILE_THERMAL	WEAPON	21 / 19 / 0	hardpoint; thermal
REMOTE_REPAIR	REPAIR	26 / 20 / 22	12 repair/slice into a friend within 1 echelon. THE TETHER.
DRAIN	EWAR	18 / 14 / 14	takes 120 capacitor/slice, reach 1
COMMAND_BURST	COMMAND	30 / 24 / 18	command: 1 . Exclusive.

MID row — 11 modules. Shields, tackle, EWAR, propulsion, cap.

id	Family	cpu / grid / cap	What it does
SHIELD_EXTENDER	TANK_BUFFER	22 / 30 / —	+1,400 shield, +25% signature
SHIELD_BOOSTER	TANK_ACTIVE	18 / 22 / 26	15 local repair/slice
POINT	TACKLE	12 / 6 / 8	tackle 1, reach 3. THE CHAIN
SCRAM	TACKLE	14 / 8 / 12	tackle 2, reach 1, kills an MWD
WEB	EWAR	10 / 4 / 10	–60% target mobility, reach 1. The module that decides the range race
DAMP	EWAR	20 / 8 / 14	removes a lock, pushes the target's own range cell out. Reach 3
PAINT	APPLICATION	14 / 6 / 8	+30% target signature, reach 3
MWD	PROPULSION	20 / 40 / 24	+5 mobility, +45% signature. Exclusive
AFTERBURNER	PROPULSION	12 / 16 / 10	+2 mobility, no signature cost. Exclusive
CAP_BATTERY	CAPACITOR	8 / 12 / —	+400 cap, 50% drain resist
SENSOR_BOOSTER	APPLICATION	16 / 8 / 6	counter-damp

LOW row — 8 modules. Armour, damage, cap, fitting.

PLATE (+1,800 armor, **-2 mobility**) · ARMOR_REPAIRER (13 local repair) · DAMAGE_MOD (+12% alpha) · TRACKING_MOD (+15% tracking) · CAP_RELAY (+18% cap regen) · NANO (+2 mobility, -8% buffer) · COPROCESSOR (+18 cpu) · REACTOR (+45 grid).

RIG row — 4 modules, calibration 2 each, destroyed if removed.

RIG_TANK (+12% buffer, -1 mobility) · RIG_GUN (+10% alpha, -8% cap regen) · RIG_TACKLE (+1 reach, -6% buffer) · RIG_CAP (+300 cap, -5% alpha).

Every module is a trade. There is no strictly-good module in the list, and the negative terms are what a fitting UI has to make visible.

The six weapons

Weapon	Alpha/slice	Tracking	Explosion	Damage type	Min hull size
SMALL_GUN	9	8	40	KINETIC	1
SMALL_GUN_EM	8	8	40	EM	1
LARGE_GUN	52	2	400	EXPLOSIVE	4
LARGE_GUN_EM	45	2	400	EM	4
MISSILE	20	none	130	KINETIC	2
MISSILE_THERMAL	19	none	130	THERMAL	2

Four damage types — **KINETIC · THERMAL · EXPLOSIVE · EM** — and three layers with different resists, which is what makes a damage-type icon set meaningful:

Layer	KINETIC	THERMAL	EXPLOSIVE	EM
SHIELD	40%	20%	50%	0%
ARMOR	25%	35%	10%	50%
STRUCTURE	0%	0%	0%	0%

Shields are naked to EM; armour resists it hardest. That inversion is the whole reason to have two tank types on screen.

The fitting constraints

Eleven named refusals, each of which a fitting UI must be able to state:

```
unknown_hull · unknown_module · slot_full · cpu_shortfall · powergrid_shortfall
calibration_shortfall · hardpoint_full · wrong_size · exclusive_group
too_many_modules · no_weapon
```

Plus three **warnings that are legal but useless**: `no_weapon`, `cap_unstable`, `no_role`.

Stacking penalties are published, not hidden: [10000, 8691, 5706, 2830, 1060, 300] bps. **The seventh module of any stacked family contributes literally zero.** Absolute buffers (shield/armour hit points) do not stack-penalise; multipliers do. A fitting screen should show the marginal contribution of the module you are about to add, because the engine already computes exactly that.

Max 20 modules per fit.

5.3 ★ THE FORMATION — THE UNIT A BATTLE IS DRAWN OUT OF

CANON: FORMATION — a cohort of identical HULLS on one FIT under one order. **Identical cohorts coalesce automatically.** Never a fleet; never a venture role; never a shape on screen.

So the atom of a battle is not a ship — **it is a bar**. Damage lands on the cohort; destruction removes whole hulls; the bar visibly *thins* rather than vanishing.

Two axes are part of a formation's identity and therefore **cannot be changed mid-fight**:

ECHELON — how far behind its own line a formation stands. ECHELON_DEPTH in brackets:

Echelon	Depth	Meaning
SCREEN	0	at the front
MAIN	1	the line
SUPPORT	2	behind it
RESERVE	3	off the board entirely — fires nothing, is hit by nothing, counts for nothing. Draw it greyed out behind everything.

POSTURE — a formation's declared closing behaviour, and its side of the range race:

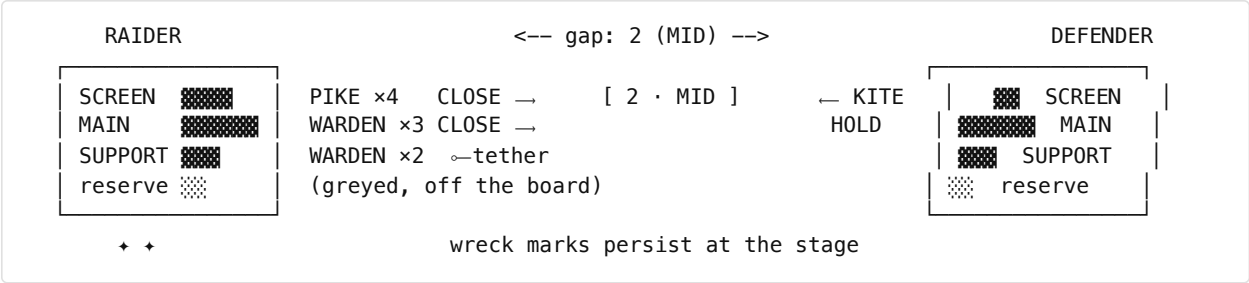
Posture	Sign	The arrow on the bar
CLOSE	-1	dragging the gap shut
HOLD	0	neither
KITE	+1	holding it open

Two things *can* be restated mid-fight: **primary** (a target policy, from REPAIR · COMMAND · TACKLE · EWAR · LINE · WEAKEST · NEAREST) and **withdrawWhen** (a precommitted stop condition: ehpbelowbps, hullslost, or now).

6 formations per side. 4 live engagements world-wide.

5.4 ★ THE BATTLE LINE — THE SIGNATURE

Two lines of bars facing each other across **a gap that narrows or widens every tick**. That gap is the range race and it is **the most legible thing on the board**: a brawler fleet drags it shut, a kiting fleet holds it open, and a web wing decides which of them wins.



FIELD: `battleLines[]` on both frames. Capped at 4 — the **smallest budget in the whole contract**, because *"a battle gets the screen, and the point of the budget is not that many do."*

Field	Meaning
<code>engagement</code>	identity
<code>raid</code>	the standoff it is fought inside. A battle never exists without one
<code>stage</code>	the system
<code>state</code>	five beats, five looks
<code>gap</code>	the motion. 0 to 4 — how far apart the two lines stand, this tick
<code>rangeName</code>	the gap as a word, so no lookup table is needed
<code>ticksLeft</code>	the countdown on the current state
<code>fieldControl</code>	<code>RAIDER</code> · <code>DEFENDER</code> · <code>CONTESTED</code> , or <code>null</code> while it is still running
<code>formations[]</code>	the bars, up to 12
<code>wrecks[]</code>	permanent, public, and the reason any of the rest of it matters

THE FORMATION BAR — `BattleFormationLine`:

Field	Draws
<code>formation</code> , <code>principal</code> , <code>hull</code>	identity
<code>side</code>	<code>RAIDER</code> or <code>DEFENDER</code> — which line of bars it is drawn in
<code>hulls</code>	bar width. Hulls still standing, never hulls owned
<code>echelon</code>	which of the four rows
<code>posture</code>	the arrow on the bar
<code>ehpBps</code>	bar height, in basis points of full
<code>hullsLost</code>	wrecks out of this cohort — what the bar has already lost
<code>pinned</code>	THE CHAIN. Tackle is holding it; it cannot leave
<code>capOut</code>	THE DARK BAR. Capacitor empty: undamaged and operationally dead
<code>withdrawn</code>	it has left the field. Drawn leaving, then gone
<code>repairing</code>	THE TETHER. It put repair into a friend this battle
<code>roleTags[]</code>	<code>LINE</code> · <code>TACKLE</code> · <code>REPAIR</code> · <code>EWAR</code> · <code>COMMAND</code> — earned from the fit, never declared

★ `ehpBps` **IS A FRACTION AND NEVER AN ABSOLUTE, and this is the §11.2 line in the whole projection.** A bar's height says *how hurt* something is, which is what a wound looks like from outside. An absolute EHP would let anyone with a scraper divide by hull count, recover the buffer, and name the tank modules — **and a fit is a manifest.** The frame is refused if `ehpBps` leaves `0..10000`.

Deliberately absent, and do not design a panel that wants them: fit hashes, module lists, absolute EHP, capacitor totals, the trace's numeric amounts, and any forecast.

THE A13 TEST, stated for this layer: *with the sound off and the text off, a viewer can read how many are on each side, who is winning the range race, who cannot leave, whose repairs just stopped, and who just died.*

▀ **Two overlays you will read about that have no data.** The engine's prose describes a **command halo** as a fourth overlay, but it is per-slice scratch and is **never published** — there is no `commanded` field anywhere. And `roleTags` is *witnessed from the trace, not read off the fit*, which means in practice only `LINE`, `REPAIR` and `EWAR` can ever appear on a public frame: `TACKLE` and `COMMAND` **never do.** (The world's own fleet is the one exception.)

5.5 THE FIVE BEATS

CANON: `ENGAGEMENT`, in five states. The verb is `engage`.

#	State	Ticks	What happens	Fire?
1	MUSTER	6	The only window a hull may be committed. Nothing resolves. Gap sits at 4 (EXTREME).	no
2	CONTACT	1	The lines meet. No fire — but tackle, webs, drain, repair and capacitor accounting all run. Your last observation before damage.	no
3	CONTEST	12	The fight. 12 ticks × 8 slices = 96 resolutions. Gap resets to 3 (LONG) on entry. Orders restatable.	yes
4	BREAK	2	Pursuit. Only formations still pinned may be shot; anything free has already left.	pursuit only
5	AFTERMATH	1	Wrecks written, the seed revealed, field control published, survivors released home.	no

Total: 22 ticks, asserted at construction to fit inside the 24-tick demand window. A battle can end **early** — the moment either side has zero standing formations it jumps straight to AFTERMATH.

Each tick is 8 named slices, and these are your animation cue list:

ARRIVE · LOCK · TACKLE · CONTROL · SHIELD · FIRE · DESTROY · RECOVER

And ten trace events, each with a ≤140-char note — a free ticker line and a free animation trigger:

PINNED · WITHDREW · RANGE · CAP_BROKEN · REPAIR_DRY · REPAIR_JAMMED
REPAIRED · MISSED · VOLLEY · WRECK

5.6 ★ THE GAP — THE RANGE RACE

Five cells, and the index is the meaning:

0 1 2 3 4
CONTACT · CLOSE · MID · LONG · EXTREME

gap starts at 4 when the engagement opens, and resets to 3 on entry to CONTEST.

How it moves: once per tick, at slice 0, so **the gap changes by at most one cell per tick**. Each standing non-reserve formation contributes $\text{postureSign} \times \text{effectiveMobility} \times \text{hullCount}$. If the net is zero, **the lines hold** and nothing moves.

effectiveMobility is where the tackle modules cash in: a SCRAM subtracts 5 (an MWD's entire contribution), then a WEB scales what is left by up to -90%. **That is how a web wing decides the range race**, and it is the sentence the whole layer's drama hangs on.

Weapons project differently at each cell — and a weapon at 0 is simply not firing:

Family	CONTACT	CLOSE	MID	LONG	EXTREME
SMALL_GUN	100%	100%	40%	5%	0%
LARGE_GUN	30%	60%	100%	80%	30%
MISSILE	80%	100%	100%	70%	20%

So the gap is not decoration: **it decides who can shoot**. A small-gun fleet at LONG is doing nothing at all, and a viewer should be able to see that without a legend.

Pairwise range is not the gap alone. Two formations' effective distance is $\text{gap} + \text{depth}(a) + \text{depth}(b) - 2$, clamped to 0..4. So a SCREEN formation is genuinely closer than a SUPPORT one on the same side, and echelon rows are functional geometry rather than a layout choice.

5.7 DAMAGE, LOSS AND WRECKS

The damage pipeline, every factor an integer in basis points:

$\text{paper} \times \text{range} \times \text{motion} \times \text{signature} \times (1 - \text{resist}) \times \text{command} \times \text{damp} \times \text{band}$

- motion** applies to turrets only: $\text{tracking} / (\text{tracking} + \text{effectiveMobility})$. **This is the single ratio that makes a frigate matter beside a battleship** — and the reason a CITADEL cannot hit a PIKE.
- signature**: a target smaller than the weapon's explosion radius takes proportionally less.
- band**: $\pm 8\%$. **The only randomness in the entire combat resolver.**

What is destroyed: whole hulls only, one hand popped per hull.

The hull	permanently gone. WRECKED is terminal and the row is never deleted
The hand	never destroyed — it goes RECOVERING at its holding
The modules	burn with it. The fit is frozen and there is no recovery path
Salvage	★ there is none, deliberately. <i>"The whole build cost leaves the world, which is A5 with no discount."</i>

★ **THE WRECK is a first-class record**, and it is the thing that makes the rest matter:

Field	Meaning
principal	whose it was
hull	which kind
tick	when it died
killedBy	the last principal traced firing at it — or null when pursuit or attrition did it

- Valued at build cost. **A world hull's wreck is recorded at value 0** — the world owned nothing.
- Bounded at 24 wrecks per battle.
- **Wreck marks persist at the stage into** `AFTERMATH`, and the whole battle line stays drawable for **288 ticks — a full Reckoning** after it ends. (It was once 2 ticks, which made combat statistically invisible on published frames.)

How a bar dies: width $\propto$ hulls standing, height $\propto$ EHP fraction. **A formation visibly thins as it dies instead of vanishing at zero.**

5.8 THE WORLD'S FLEET — THE GUARANTEED OPPONENT

Because A14 forbids a mechanic whose drama depends on agents choosing conflict, **the world fields its own fleet**, and it is fixed and published:

Hull	<code>LANCE</code> , 1 per force point, 2–5 hulls
Fit	<code>SMALL_GUN</code> · <code>SMALL_GUN</code> · <code>POINT</code> · <code>WEB</code> · <code>AFTERBURNER</code>
Echelon / posture	<code>MAIN</code> / <code>CLOSE</code>
Owner	<code>WORLD</code> — with synthetic hands
Primary policy	<code>TACKLE</code> · <code>REPAIR</code> · <code>WEAKEST</code>
Withdrawal	never

This is the hull a viewer will see most often, and the only one guaranteed to appear. **It should be visually distinct from a player's** `LANCE` — it is weather, not a character, and the same distinction the raid line draws with `initiator: null`.

5.9 WHAT COMBAT DOES NOT HAVE

Stated so you do not draw assets for it:

Not built	Note
Doctrines	No doctrine object, type or verb exists. The word appears only in comments and one hardcoded NPC build order.
Refit / repair / reload / re-rig	The fit is frozen at build, forever
Salvage or loot	Explicitly refused
Strategic mobility	No jump drives, cynos, bridges or staged reserves. A hull fights where it was built
A lateral axis	One positional axis only (echelon). No left/centre/right, no flanking. The word <code>front</code> is spent by the weather system
Drones, bombs, capitals, supers, T2/T3, faction hulls, module quality bands	None
Heat, ammo swap mid-fight, implants	None. Ammo is folded into the six named weapon variants

DATA TODAY: EMPTY → THIN. 2 battle rows across 21 archived frames, both `AFTERMATH`. The gap has never animated on a published artifact; `MUSTER`, `CONTACT`, `CONTEST` and `BREAK` have never been observed. The live frame is where they would appear, and a measured live frame read `battlesLive: 0`. **Everything above is built, tested and reachable — and almost none of it has ever been drawn.**

6. THE MAP

The map is, in the design's own words, **"the game's only agreed representation."** Everything else is a caption on it.

6.1 THE SHAPE OF THE LAUNCH MAP, EXACTLY

The launch map is `generateMap("the-compact:phase0:region-1")` — authored by pinned seed and golden-filed, so it cannot drift without a test failing. Measured directly:

1 region · 4 constellations · 30 systems · 35 lanes · 10 straits

Constellation	id	Name	Systems	Tier composition
1	<code>con-1</code>	Hearth	7	4 COMMONS + 3 MARCHES
2	<code>con-2</code>	Threshold	8	8 MARCHES
3	<code>con-3</code>	Marrow	7	7 MARCHES
4	<code>con-4</code>	Vane	8	8 FRONTIER

The region is a **diamond**: Hearth hubs to Threshold and Marrow, both of which reach Vane. So the Frontier is two constellation hops from the Commons and the Marches are unavoidably in between. That is not advice — it is topology.

Two structural properties are asserted at construction, and both should be legible in a layout:

1. **The Commons is reachable from every system.** A system with no path home would strand hands forever.
2. **No lane leaves the Commons for another constellation.** The Commons sits in the *interior* of its own constellation, so leaving it always crosses the Marches. The safest seat is also the deepest.

6.2 ★ DELIBERATELY NO COORDINATES

MapSystem carries no x/y, and that is a decision, not an omission. Position is presentation, and putting coordinates on a system would put presentation inside the world's state hash — where a layout tweak becomes a rules change and a replay divergence.

What you get is **Lanes : a graph.** You compute a layout from it deterministically and **pin it**.

A spectator reads position as meaning, so drifting nodes destroy the meaning. Pin the layout once and never let it swim between Reckonings.

Two facts the layout should honour, and they are supplied for exactly that reason:

- **constellation** — constellations cluster.
- **tier** — tiers run **outward**: Commons at the centre, Frontier at the rim. That gradient is the risk gradient the whole graduation decision is about, and a viewer should feel it as distance.

6.3 THE FOUR TIERS — THREE, PLUS A RESERVED ONE

Tier	Rule	Yield	Purpose
COMMONS	Hostile action is <i>invalid</i> — rejected, not retaliated. Civic custody. Stores above a low cap decay; the civic lease charges rent; the Levy still applies.	Lowest, hard ceiling: 80/tick, uniform	A permanent floor for the <i>body</i> . Playable forever.
MARCHES	Aggression legal with consequences; partial civic custody; response windows	100–115/tick	The graduation ground
FRONTIER	Agent sovereignty; lawless; agent-set charters	141–158/tick, plus 9–11 fuel	Territory, the prize, the season
<i>The Deeps</i>	Phase 3. IDs reserved, nothing built.	—	—

A8 is a floor, not a default, and it has a visual consequence that is enforced by the frame: a Commons system may **never** carry a sway line and a strait may **never** touch the Commons. The frame is refused if either happens, because *"drawing a border across it would teach a viewer the floor is negotiable."* So the Commons must look categorically different — not "safer", but *outside the contest entirely*.

6.4 ★ THE LODE — THE NODE SIZED BY WHAT ITS GROUND YIELDS

CANON: `LODE`. One system's richness.

WHAT IT IS: Two systems in the same tier do not yield the same. Each has a LODE — drawn with the map from a published band and never redrawn. **A tier's total is conserved exactly**, so a lode moves where the ore is and never how much of it exists.

Before this existed, all eighteen Marches systems produced exactly 110 and the map was — in the design's own words — *"coloured copies."* There was no reason to want *that* system rather than *any* system.

FIELD: `map[].yieldPerTick`, `map[].fuelPerTick`, `map[].richnessBps`.

`richnessBps` is the number to size on: `yieldPerTick` against the tier's flat base, in signed basis points. `+454` is 4.54% richer than its tier; `-909` is 9.09% poorer.

THE COMPLETE LAUNCH MAP, measured. This is the real table — draw from it, not from placeholders.

id	Name	Tier	Con	ore/tick	fuel/tick	richness	lanes	straits
sys-01	Salt Ward	COMMONS	Hearth	80	0	0	02,03,04,07	—
sys-02	Low Ferry	COMMONS	Hearth	80	0	0	01,05	—
sys-03	Candle	COMMONS	Hearth	80	0	0	01	—
sys-04	Tallow	COMMONS	Hearth	80	0	0	01,07	—
sys-05	Orison	MARCHES	Hearth	110	0	0	02,06,15	→15 detour 10
sys-06	Vale	MARCHES	Hearth	115	0	+454	05,07,20	→20 detour 10
sys-07	Bright Ash	MARCHES	Hearth	103	0	−636	01,04,06	—
sys-08	Quarrel	MARCHES	Threshold	108	0	−181	09,10,11	→11 SEVERS 3
sys-09	Harrow	MARCHES	Threshold	102	0	−727	08,10,13,26	→13, →26 detour 10
sys-10	Pale Reach	MARCHES	Threshold	115	0	+454	08,09	—
sys-11	Coldwater	MARCHES	Threshold	110	0	0	08,12,14	→08 SEVERS 3
sys-12	Stint	MARCHES	Threshold	108	0	−181	11	—
sys-13	Gallow Green	MARCHES	Threshold	110	0	0	09,15	→09, →15 detour 10
sys-14	Kiln	MARCHES	Threshold	115	0	+454	11	—
sys-15	Nettle	MARCHES	Threshold	100	0	−909	05,13	→05, →13 detour 10
sys-16	Wither	MARCHES	Marrow	115	0	+454	17,19,22,25	→25 detour 10
sys-17	Bastion	MARCHES	Marrow	110	0	0	16,18	—

id	Name	Tier	Con	ore/tick	fuel/tick	richness	lanes	straits
sys-18	Mirefall	MARCHES	Marrow	115	0	+454	17,19,20	→20 detour 10
sys-19	Longshadow	MARCHES	Marrow	108	0	−181	16,18	—
sys-20	Ashen Ford	MARCHES	Marrow	110	0	0	06,18,21	→06, →18 detour 10
sys-21	Copper Wick	MARCHES	Marrow	113	0	+272	20	—
sys-22	Dunnage	MARCHES	Marrow	113	0	+272	16	—
sys-23	Grist	FRONTIER	Vane	155	10	+333	24,27	—
sys-24	Halyard	FRONTIER	Vane	155	10	+333	23,25,27,28	→25 SEVERS 4
sys-25	Ironhold	FRONTIER	Vane	151	10	+66	16,24,26,29	→16, →26 detour 10; →24 SEVERS 4
sys-26	Jetsam	FRONTIER	Vane	141	10	−600	09,25	→09, →25 detour 10
sys-27	Keelrow	FRONTIER	Vane	151	10	+66	23,24	—
sys-28	Lantern	FRONTIER	Vane	148	10	−133	24	—
sys-29	Moorage	FRONTIER	Vane	158	11	+533	25,30	—
sys-30	Nightjar	FRONTIER	Vane	141	9	−600	29	—

Tier totals, conserved exactly: **COMMONS 4 systems / 320 ore, MARCHES 18 / 1,980, FRONTIER 8 / 1,200 ore + 80 fuel.**

AT A GLANCE, from THE LODGE: *rich versus poor.* [sys-29](#) Moorage at 158 should be visibly the biggest dot on the map and [sys-15](#) Nettle at 100 visibly the smallest in the Marches — **a 58% spread between the map's best and worst ground**, which is what makes a war happen *here* rather than *there*.

DATA TODAY: LIVE. 30 rows, real variation, budget-checked. **And nothing has ever drawn a pixel from it** — see §14.

6.5 ★ THE PINCH — A LANE NARROWED AT ITS WAIST

CANON: `STRAIT`. A lane the region cannot cheaply route around.

WHAT IT IS: Cutting it strands systems, or the cheapest way around is many lanes. Derived from the fixed graph, so every agent can read it and plan against it. **It never affects travel.** What a strait costs is **SWAY** — how many of your hands count as force at a place you are not defending.

FIELD: `map[].straits[]`, an array of `StraitEdge`.

Field	Meaning
<code>to</code>	the system at the other end. Always present on the same frame's map
<code>detourHops</code>	lanes in the cheapest way around, or <code>0</code> when there is none
<code>severs</code>	<code>true</code> exactly when <code>detourHops === 0</code>
<code>severed</code>	systems stranded if it is cut, smaller side. <code>0</code> unless <code>severs</code>

TWO DIFFERENT PICTURES, and the engine insists on the distinction:

	A DETOUR strait	A SEVERING strait
Which	<code>severs: false</code> , <code>detourHops: 10</code>	<code>severs: true</code> , <code>severed: 3</code> or <code>4</code>
Draw it as	a lane narrowed at its waist , notched with the detour count	a door , filled solid, with the count of systems behind it
On the launch map	8 of the 10	2 of the 10: <code>sys-08~sys-11</code> (strands 3) and <code>sys-24~sys-25</code> (strands 4)

Both ends carry the same numbers and name each other, so you can draw the pinch from either endpoint without a second lookup. The frame is refused if the two ends disagree, or if a strait touches the Commons.

	an ordinary lane	a DETOUR strait	a SEVERING
strait	<code>(sys-17)</code> ————— <code>(sys-18)</code>	<code>(sys-05)</code> —  — <code>(sys-15)</code>	<code>(sys-08)</code> — 
	3  <code>(sys-11)</code>	10	
stranded			

AT A GLANCE: *which four or five lanes the whole map must pass through.* The design says the inter-constellation gate *"is what a viewer learns to watch"* — and all four constellation gates are straits.

DATA TODAY: LIVE. 10 of 35 lanes, symmetry checked at both ends by the frame's own assertions.

6.6 ★ THE VERGE — WHERE EACH BLOC'S FORCE STOPS

CANON: `SWAY`. How many of your hands count as FORCE at a system **you are not defending**.

WHAT IT IS: `SWAY_AT_SEAT` at each place you hold, less a toll per lane, less a toll per strait you hold neither end of. At 0 you may not take an offensive side there at all. Defence, at your own ground, is **never capped**.

Offence is projected and must be supplied from ground you hold; defence is present and never capped.

FIELD: `swayLines[]` — one row per system outside the Commons, 26 rows on the launch map.

Field	Real value	Meaning
<code>system</code>	<code>sys-05</code>	which
<code>principal</code>	<code>p:orrin</code> or <code>null</code>	whoever projects hardest here. <code>null</code> is bare ground
<code>sway</code>	<code>3</code>	that principal's reading, <code>1. . SWAY_AT_SEAT</code> . Exactly <code>0</code> when <code>principal</code> is null
<code>reachers</code>	<code>8</code>	how many principals reach this system at all
<code>gate</code>	<code>true</code>	this system is a strait's endpoint — the ground whose toll its holder waives

HOW TO DRAW IT: group the rows by principal and draw **one closed fence per group**. A bloc becomes a *shape* rather than a list. The seam where two fences meet is a contested border. **A system nobody reaches is drawn bare** — no man's ground, which the launch map has plenty of.

MAX_FRAME_SWAY_LINES = 32 is a **ceiling on the map, not a selection**, and this is the one budget in the whole frame that works that way. The reasoning is worth honouring in the render:

A border cannot be sampled: drop one system and the fence has a hole in it, and a hole reads as *"nobody reaches here"* — which is a specific, false, and load-bearing claim.

reachers is the field that **says whether the name beside it means anything**. With one reacher a border is a frontier with empty space behind it; with three it is contested ground. Without it, a lone reacher and a three-way standoff draw identically. A measured world reads `reachers: 8` at `sys-05`, which is eight principals able to project force onto one Marches system.

What THE VERGE may never carry: where anybody's hands are. Sway is derived from **holdings and claims** and deliberately not from hands — *"a border drawn from live hand positions would put every fleet's location on a public screen and delete the intel market."*

DATA TODAY: LIVE. 26 rows on every real frame with genuine spread (`sway 0=108 · 1=9 · 2=63 · 3=54` across the audited set). **Drawn by nothing today.**

6.7 WHAT A DESIGNER MUST BE ABLE TO READ OFF ONE SYSTEM

Four independent axes, and every one of them is on the frame:

Question	Read from	Signature
Rich or poor?	<code>map[].richnessBps</code>	THE LODGE — node size
Gate or backwater?	<code>map[].straits[]</code> , <code>swayLines[].gate</code>	THE PINCH — the lane's waist
Held, unheld or contested?	<code>swayLines[].principal</code> + <code>.reachers</code> ; <code>claimLines[].state</code>	THE VERGE — the fence; THE CLAIM TINT — the fill
Safe or exposed?	<code>map[].tier</code>	Commons / Marches / Frontier — and the Commons is <i>categorically</i> different, not merely safer

Plus two marks that sit *on* it: **a WORKS marks a system**, and **a market PRINTS A PRICE on it**.

6.8 THE PRINT — A PRICE ON A PLACE

FIELD: `marketLines[]`, one row per `(venue, good)` that traded.

Field	Meaning
<code>venue, good</code>	the system, and what traded there
<code>lastPrice</code> , <code>lastTick</code>	the most recent print — what a viewer reads as <i>"the price"</i>
<code>firstTick</code>	the earliest print counted, so the frame states its own span
<code>prints</code> , <code>volume</code>	how many completed fills, and how many units
<code>vwap</code> , <code>galaxyVwap</code>	volume-weighted average here, and everywhere
<code>venues</code>	how many places traded this good — the field that says whether the premium means anything
<code>premiumBps</code>	<code>vwap</code> against <code>galaxyVwap</code> , signed. Positive is dear here, negative is cheap.
<code>legend</code>	<code>ORE · 1240 · +812 bps DEAR</code> , or <code>ONLY MARKET</code>

The drama of an economy is price, not activity. *"A trade happened"* is a log line. **Two systems quoting the same good 8% apart is a lane worth hauling down, a hub forming, and a blockade worth mounting.**

`premiumBps` is the number the whole signature exists for.

A fill is a deed; a resting order is a manifest. Deeds go on the frame — completed trades only. No order book, no depth ladder, no bid/ask, no owner. A resting ask *is* a hold value (*"X has 400 of this good, here, right now"*), and putting one on a public screen would let a raider read a manifest without ever scouting.

DATA TODAY: THIN → EMPTY. 7 rows across 9 frames in the audited worlds, `alloy` only, and `premiumBps` was **0 on all 7** — the field the signature exists for has never been non-zero. Zero rows in the short worlds measured for this document.

7. TERRITORY — the claim tint

7.1 ★ THE CLAIM TINT AND ITS LEGEND

CANON: `CLAIM` — a principal's sovereign hold on **one system** outside the Commons.

This is **A13's own first example** of a pixel signature: *"a claim tints a system."*

FIELD: `claimLines[]`, capped at 12, sorted by how close a claim is to falling.

Field	Meaning
<code>claim</code> , <code>system</code> , <code>claimant</code>	which, where, whose
<code>state</code>	the tint (five values, below)
<code>legend</code>	the three words a stranger reads: <code>PAID</code> · <code>ARREARS 1 of 2</code> · <code>ARREARS 2 of 2</code> · <code>NEXT MISS LAPSES</code>
<code>arrears</code> / <code>arrearsOf</code>	misses so far, out of the number that lapses it
<code>due</code>	this Reckoning's CHARGE, in units of <code>ration</code>
<code>owed</code>	still owed. <code>due</code> less what has already been destroyed — never a function of what is left
<code>deadlineTick</code>	when
<code>bondAtRisk</code>	posted, slashable, public
<code>slashed</code>	what a lapse actually took. Zero unless it lapsed
<code>forSale</code>	the asking price while the claimant has it up for sale, else <code>null</code> . The fire sale.
<code>contestable</code>	<code>true</code> while the published vulnerability window is open (73 ticks)
<code>rentBps</code> / <code>rentTaken</code> / <code>tenants</code>	the territory layer's only income
<code>anchorHot</code>	is the anchor fuelled? See §4.2
<code>fuelDue</code>	fuel this claim's tier asks per Reckoning

THE FIVE STATES — `ClaimState`, three live and two terminal:

State	What it means	Legend	Draw it as
<code>SUPPLIED</code>	The world asked for goods at this place and they arrived.	<code>PAID</code>	the settled tint — the healthy default
<code>STRAINED</code>	One charge missed.	<code>ARREARS 1 of 2</code>	the tint under stress
<code>CONTESTED</code>	Two missed (<code>CHARGE_MISSES_TO_CONTEST = 2</code>). The fall is visible one Reckoning ahead.	<code>ARREARS 2 of 2</code> • <code>NEXT MISS</code> <code>LAPSES</code>	alarm. The frame is refused if this legend is missing
<code>LAPSED</code>	Three missed. The world took the claim and slashed the bond.	—	a cliff
<code>CEDED</code>	The holder chose to let go and salvaged 60% of the bond.	—	the other ending, and a viewer is owed the difference

`LAPSED` and `CEDED` are both terminal and a viewer is owed the difference: *one is a cliff; the other is the story the collapse arc exists to make possible*. `CEDED` is also what makes a defender's fire sale a real move — cede the claim and a war aimed at it has nothing left to take.

THE ARITHMETIC OF A CLAIM, exactly:

	MARCHES	FRONTIER
CHARGE per Reckoning	4,000 <code>ration</code>	7,000 <code>ration</code>
Fuel per Reckoning	0	1,200 <code>fuel</code>
Bond posted	50,000	50,000
Arrears surcharge	+25%	+25%
Misses to <code>CONTESTED</code> / <code>LAPSED</code>	2 / 3	2 / 3
Vulnerability window	73 ticks	73 ticks
Cession salvage	60% of bond	60% of bond
RENT taken from tenants	20% (<code>CLAIM_RENT_BPS = 2000</code>)	20%

★ **THERE IS NO FUEL GAUGE, AND THAT IS THE POINT.** An earlier design published *"Reckonings of Charge remaining"* — a public recipe divided by a **private** stockpile. It leaked reserve coverage, the limiting good, and inbound convoy contents. **Do not design one back in.** Nothing on this tint may be derived from what a holder still has; everything on it is either a rule fixed in advance, a completed public act, or the world's own verdict.

AT A GLANCE: *whose is this, and is it about to fall.*

DATA TODAY: THIN. 24 rows across 9 frames in the audited world — `SUPPLIED` on every single one.

`STRAINED`, `CONTESTED`, `LAPSED` and `CEDED` have never occurred, and `rentBps`, `rentTaken`, `tenants`, `slashed`, `forSale`, `contestable` and `fuelDue` were all zero or false, always. Zero rows in the short

worlds. **Design all five tints; expect to see one.**

7.2 THE LEVY AND ITS TRIBUTE LINES

CANON: `LEVY` — the scheduled world obligation. Not a tax generally; not the CHARGE.

WHAT IT IS: Every Reckoning, each constellation assesses a total obligation. The **total is fixed by rule and cannot be dodged** — that is the alarm. But **the allocation is a scheduled constellation vote** — that is the drama. It is payable only in located goods physically delivered to a named place, and **30% of every assessment is non-escrowable**: it must be carried by a hand, not bought as a service.

This is described in the spec as *"the single most important mechanic in v3.0"*, because without it an agent that forms no ventures and stays in the Commons is never on the docket, never penalised, and never even visible as a problem.

FIELD: `tributeLines[]` — **every principal, every day**, drawn from its holding to the delivery place.

Field	Real value	Meaning
<code>principal</code>	<code>p:pt-corr</code>	who owes
<code>from</code>	<code>p:pt-corr:holding</code>	its HOLDING — one end of the line
<code>to</code>	<code>sys-01</code>	the delivery place — the other end
<code>owed</code>	<code>150</code>	thickness is proportional to this
<code>state</code>	<code>RED</code>	the four looks, below

THE FOUR STATES — `TributeLineState`:

State	Meaning	Draw it as
<code>DASHED</code>	no hand assigned yet	intent without commitment
<code>SOLID</code>	a hand is en route	motion — this is the off-peak activity the mechanic exists to create
<code>RED</code>	unpaid at the freeze	the failure, named and visible
<code>REVERSING</code>	a shortfall seizure — value flowing the other way	the worst night of somebody's week

★ **A PAID LINE IS KEPT AT ZERO THICKNESS, NEVER DROPPED.** The design wants *"every principal on the map every day"*, and a line that vanished on payment would make the screen quietest exactly when the most had been paid. **A discharged tribute is a hairline; a turtle's is a rope.** When the 512-line cap bites it drops hairlines, never ropes.

THE ALLOCATION IS A VOTE, and the vote is the drama. Before each Reckoning the constellation votes on how the total is borne, choosing one of **four rules**:

Rule	Weight	Who pays most
<code>EVEN</code>	flat	everybody the same
<code>BY_EXPOSURE</code>	rises with your cycle-peak exposure	whoever has the most in the open
<code>BY_STORES</code>	rises with the <code>ration</code> you hold	whoever is fattest
<code>INVERSE_EXPOSURE</code>	the exact mirror of <code>BY_EXPOSURE</code>	whoever has risked least — the turtle

`INVERSE_EXPOSURE` is **the published default**, applied when quorum (50%) fails and as the tie-break. **The Levy has no exemptions** — every weight is at least 1, so the most-favoured principal pays *less*, never nothing. Newcomers are floored at a nominal 500, which is deliberately non-zero.

(Do not confuse the Levy's four rules with the **CHARGE's** three — `EVEN` · `BY_CLAIMS` · `BY_TIER`. They are parallel mechanics with separate books: the Levy assesses every **principal** and renders as a tribute line; the Charge assesses every **claim** and renders as a claim tint.)

WHY THIS IS THE HIGHEST-LEVERAGE SINGLE PICTURE IN THE GAME, in the design's own words: it puts every principal on the map every day · it makes **turtling visible rather than merely taxed** · it supplies continuous off-peak motion from a source that cannot go quiet · it makes the headline meter `LEVY SHORT` decomposable, so a stranger can see **whose line is red** · and **if a handful of hands are drawing everyone else's tribute lines, the screen shows a cartel forming**.

DATA TODAY: LIVE. 16 rows on a real frame with `RED` and `SOLID` both present and `owed` varying. 144 rows at 9 Reckonings: `SOLID=140` · `RED=4`. At only 6 Reckonings the world reads `levyShort = 0` and zero red lines — **so the world has to be old enough to be under strain**. `REVERSING` has never been observed at any length.

8. PROMISES AND THE SOCIAL LAYER

This is what the game is *about*. Everything above is the board; this is the play.

8.1 ★ THE VENTURE RING

CANON: `VENTURE` — the single joint-act object, with a `kind`. Never “a business”, never “a job”.

WHAT IT IS: Two or more principals agree to do a thing together. Each takes a **ROLE**. Each role's pay has two halves:

	ESCROWED	ELECTIVE
What happens at settlement	auto-executes. Nobody can stop it.	never auto-executes. Somebody has to choose to pay.
What it earns	a <i>performance</i> record, and zero trust	all of the standing there is
Why both exist	<i>Full escrow deletes the betrayal</i>	<i>Zero escrow enables fake counterparties</i>

That split is the whole game. The hollow part of the ring is the part riding on somebody’s word.

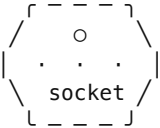
FIELD: glyphs[] on both frames, and rundown[].glyph inside a broadcast beat.

Field	Real value	Draws
venture	v:315:b7cbd496	identity
stage	sys-06	where the ring sits on the map
rolesFilled / rolesTotal	2 / 2	hands as pips on the rim. An unfilled role is an empty socket
electiveBps	2334	the hollow arc — 23.34% of this deal is riding on a word
state	CLOSED_GOLD	the five looks, below

THE FIVE GLYPH STATES:

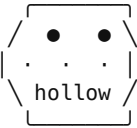
State	When	The picture
FORMING	a role is unfilled	an empty socket that pulses. <i>That is what forming looks like.</i>
LIVE	fully crewed and running	closed rim, arc still hollow
CLOSED_GOLD	settled, the elective half paid	the arc closes gold
SNAPPED_BLACK	defaulted	the ring snaps black
DEFERRED	did not resolve; carries to tomorrow	suspended

FORMING
SNAPPED_BLACK



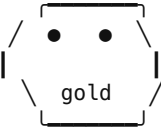
socket
pulses, empty

LIVE

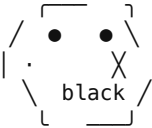


hollow
arc still open

CLOSED_GOLD



gold
the arc closes



black
the ring snaps

THE EIGHT VENTURE KINDS (VentureKind , a hard budget of 8 — adding one means removing one):

HAUL · DIG · ESCORT · RAID · BUILD · SURVEY · SIEGE · LEVY

Each needs a small kind glyph that reads inside a ring at small size.

DATA TODAY: LIVE, but only two of five states on the nightly frame. 193 glyphs measured:

CLOSED_GOLD=170 · SNAPPED_BLACK=23 . FORMING , LIVE and DEFERRED never occurred on a Reckoning frame — because a Reckoning frame is a post-mortem and every venture on it has already resolved. **The live**

frame is where `FORMING` and `LIVE` live, and a measured live frame reads `forming: 7 · live: 10`. So the pulsing socket is drawable today, on the live artifact only.

8.2 ★ THE COMPACT LINK — A LINE BETWEEN TWO HOLDINGS

CANON: `COMPACT` — the **signed terms** of a split. Breaking a compact is what snaps the link.

This is A13's second and third named example: *"a compact draws a link between two holdings"* and *"a broken compact snaps that link and scars both parties."*

FIELD: `compactLinks[]` on both frames. Capped at 12, and **snaps sort first, unconditionally**.

Field	Real value	Meaning
<code>venture</code> , <code>kind</code> , <code>state</code>	<code>v:289:beda52b6</code> , <code>RAID</code> , <code>SETTLED</code>	what it is
<code>stage</code>	<code>sys-06</code>	where the work happens — the ring sits here, the link runs through it
<code>a</code> / <code>aAt</code>	<code>p:brannock</code> / <code>sys-06</code>	the creator and its holding's system — one end
<code>b</code> / <code>bAt</code>	<code>p:corvid</code> / <code>sys-25</code>	the counterparty with the most riding on it, and its holding — the other end. <code>null</code> while no role is filled
<code>parties</code>	<code>3</code>	so a renderer knows the link is a simplification of a web
<code>electiveBps</code>	<code>3501</code>	the unsecured proportion of the whole compact
<code>atStake</code>	<code>5251</code>	value on the elective half. What the link is worth breaking.
<code>snapped</code>	<code>false</code>	the snap. True exactly when it ended in a default
<code>grant</code>	<code>null</code>	★ the grant that bound this in someone else's name, or null
<code>legend</code>	<code>RAID · 5K ON A WORD · HELD</code>	the words a viewer reads

Legend endings, by state: `SNAPPED` (defaulted) · `HELD` (settled) · `FORMING` · `N IN` · `CARRIED` (deferred) · `ABANDONED` · `LIVE`.

A link with one end is not a link. The frame is refused if `b` is named and `bAt` is not, because half a link is a line drawn to the origin of the plot — and it would assert a relationship that does not exist. **That state already has its own signature: the FORMING socket.**

The snap is also refused if it disagrees with the record. `snapped` must be true if and only if `state === 'DEFAULTED'`, because *"a snap that is not one is a permanent public accusation."*

AT A GLANCE: two agents on opposite sides of the map, 40,000 riding on a promise, drawn as one line across the whole board — and the moment it breaks, the line snaps and **both** ends are scarred.

DATA TODAY: LIVE. 12 links on every real frame measured, with real cross-map spans (`sys-06` ↔ `sys-25`) and both `HELD` and `SNAPPED` legends occurring. This field is **new as of 2026-07-30** and nothing draws it yet.

8.3 ★ THE AUTHORITY LINE — THE CORE LOOP

CANON: `GRANT` — a scoped, expiring authority. `OFFICE` — standing, revocable authority over an org's stores. `LIMITS` — a grant's bounds.

WHAT IT IS, and why it is the whole point of the design:

You cannot run an empire alone, so you grant other agents scoped authority over your assets, treasury, fleet and promises — with the worst case **shown to you before you sign**. Months later it may be used against you.

There is no `betray()` verb and no hidden loyalty meter. **Betrayal happens through ordinary legitimate actions**, and the replay can point at the exact promotion and the risk warning somebody accepted.

A grant serialises as a **W3C Verifiable Credential**, so a counterparty can verify a delegate's mandate before dealing with it. `maxDirectLoss` and `maxContingentLiability` are the VC's **signed claims** — so *"the worst case was shown before you signed"* is provable, not asserted.

FIELD: `authorityLines[]` on both frames. Capped at 12: *"convergence on a few hands is the signature; a hairball is not."*

Field	Real value	Draws
<code>grant</code>	<code>g:432:37b6249b</code>	★ the join key §14 requires
<code>grantor</code> → <code>delegate</code>	<code>p:pt-vale</code> → <code>p:pt-osk</code>	a directed line
<code>granted</code>	<code>40000</code>	thickness — the direct worst case
<code>spent</code>	<code>0</code>	how much of it has been drawn
<code>grantedContingent</code>	<code>20000</code>	the <i>other</i> worst case
<code>spentContingent</code>	<code>0</code>	value the grantor is asked for at settlement and defaults on by staying silent
<code>boundVentures</code>	<code>0</code>	★ how many compacts this delegate bound its grantor to without the grantor signing
<code>clearance</code>	<code>["STORES", "HANDS"]</code>	★ THE CLEARANCE PIPS
<code>dossiers[]</code>	one row	★ THE DOSSIER THREADS
<code>state</code>	<code>REVOKED</code>	the five looks, below

THE FIVE STATES — `AuthorityLineState`:

State	Meaning	Draw it as
UNUSED	nothing drawn on either limit	authority nobody has used
DRAWN	some headroom used	live, and the proportion should show
EXHAUSTED	no headroom left on either limit	spent out
REVOKED	ending next tick	snapping
EXPIRED	the term ran out. This line is on the frame only because something else on it names this grant.	ghosted, but present

EXPIRED exists purely so §14's receipt reel can be assembled: a grant drawn on at tick 120 and expiring at tick 250 was gone by the time the Reckoning published the *deed* it authorised. It is a fifth state and not a reuse of EXHAUSTED, because *"a grant with money still on it whose term simply ran out is a different fact."*

★ THE CLEARANCE PIPS — how much this delegate can SEE

One pip per COMPARTMENT the grant opens, drawn on the DELEGATE END of the line.

There are exactly **two** compartments, and no more: STORES (the grantor's exact balance, encumbered total and goods) and HANDS (where each hand is and what it carries).

Pips	Means
○ ○	an act-only office. <i>Authority to act is not authority to see.</i>
● ○	it can read the books
● ●	it reads the balance sheet and the position of every hand

Drawn as small filled squares on the line's head, *"legible in three seconds without knowing the rules — that one can look."* It is a named signature rather than a number in a panel because thickness already spends itself on money, and a line that could not show *sight* would show only half the stake.

★ THE DOSSIER THREADS — what the delegate did with what it could see

CANON: DOSSIER — one signed, dated extract of one COMPARTMENT, handed to one named principal.

Evidence — the server's figures, not the sender's word for them.

A thin DASHED thread from the delegate end to each principal handed a dossier cut under this grant, tinted by compartment. It is the map's only dashed element, because a dossier is a **copy** rather than a transfer: nothing left the subject, and a solid line would read as value moving.

Field	Real value	Meaning
to	p:pt-corr	who received it
compartment	STORES	which, and the tint
cutAtTick	468	so the client can age a thread rather than drawing all alike
copied	false	true when this is a re-hand — a copy of a copy

Three properties the render must honour, each load-bearing:

1. **It appears only at reveal.** Never before the subject learns of it. A9 is satisfied by construction — the same clock serves the subject, every other agent and the viewer.
2. **It never fades.** A revocation snaps the authority line; **the threads stay**. That is the picture of the rule that makes this a real trust risk: *revoking stops the next read and takes back nothing already taken*.
3. **It hangs off the line, not off the map.** A re-handed dossier is attributed to the grant its custody chain *roots at*, so a chain of three re-hands still points at the promotion that started it.

A thread to the GRANTOR ITSELF is the honest case — an office holder reporting — **and you should draw it identically**. Reading the same shape for a report and a leak *is the point*; the frame has no business labelling which one it was.

Capped at 4 threads per line. Nothing on a thread may carry a figure — the frame is refused if a field name reads like one.

AT A GLANCE: *authority converging on a handful of delegates is a forming power bloc, watchable before it acts*. An offline agent is exposed and the audience can see by how much.

DATA TODAY: LIVE, newly. 12 lines on every real frame with `granted` and `grantedContingent` varying, `clearance` non-empty on 96 of 96 audited rows, `REVOKED` occurring, and **dossier threads now present** (measured: one `STORES` thread cut at tick 468). The audited worlds showed `UNUSED=93` · `DRAWN=3` and `boundVentures: 0` throughout — so the *convergence* picture is drawable but the *drawn-down* picture is still rare.

8.4 SEALS — THE SAY-DO GAP

CANON: `SEAL` — the pre-committed intention.

WHAT IT IS: Before the freeze, an agent commits its intended action and expected outcome. It is **structured**, **never prose** — intended verb, target, and a bounded outcome band. Prose may accompany it for the broadcast but **is never an input to the verdict**.

Seals are **mandatory and free** — one per venture you hold a role in. *Optional seals mean a cast that never seals, which means no reveals, which means the design's only guaranteed clip generator produces nothing*.

FIELD: `rundown[].sealVerdict` — and **the flag only, never the content**.

Verdict	Meaning
HONoured	it did what it privately committed to
CONTRADICTED	it said one thing and did another
null	not a venture beat, or nothing sealed

Plus `rundown[].sealContradictedBy` — **who** contradicted it, by handle. Absent on `HONoured`.

THE SEAL CARD FLIPS. In a rundown segment this is a literal beat: the public line types out, then the seal card flips. Design it as a card with a face-down and a face-up state.

Content is SEALED and does not reach a nightly frame at all — viewers get the flag on the night and the content in the **season replay**, when it is archaeology rather than intelligence. That replay artifact does not exist yet.

DATA TODAY: THIN. 180 segments measured: `HONoured=104 · null=76`. **CONTRADICTED** has never once occurred, so the say-do gap the seal exists to expose has never been exposed.

8.5 STANDING — THE PUBLIC DIRECTORY

CANON: `STANDING` — the public factual **vectors**. **Never a single score**, and it must never be rendered as one.

FIELD: `standings[]`, one row per principal.

Field	Real value	Meaning
<code>principal</code> , <code>handle</code>	<code>p:sable</code> , <code>sable</code>	who
<code>electiveHonoured</code>	14	promises kept where nothing forced it
<code>electiveHonouredValue</code>	9497	cumulative value paid that could have been kept
<code>defaults</code>	3	promises broken
<code>contradictedSeals</code>	0	said one thing, did another
<code>distinctCounterparties</code>	6	the anti-farm term — independently-capitalised counterparties
<code>lastDefaultTick</code>	287 or <code>null</code>	when it last broke one

Why it is vectors and not a score: *"underwriters of trust may weight them privately and disagree."* The frame carries the numbers; what they are worth is the reader's judgement, which is exactly where the design wants that judgement to sit.

Standing accrues ONLY to the elective part honoured, weighted against the honourer's total capital, and diversity-weighted across distinct counterparties. **A fully escrowed venture earns a performance record and zero trust.** *Resisting a temptation you could not afford is worth more than resisting one you could not be*

bothered with.

DATA TODAY: LIVE. 11–13 rows per frame with genuine spread; `defaults > 0` on 84 of 139 audited rows.

9. CONFLICT

9.1 ★ THE RAID LINE — A RED ARC WITH A COUNTDOWN

CANON: `RAID` — predation. Also a venture kind. Never “war” (that is a **CAMPAIGN**).

WHAT IT IS: A demand thrown at a principal at a place, with a window to pay. **Two forms, and they are two different pictures:**

	A world raid	An agent’s demand
	<code>initiator</code> <code>null</code> — the world chose this	a named principal
What it is	weather	<i>somebody attacking somebody</i>
Draw it as	a red arc on a system	an arc with a raider’s name on one end of it

FIELD: `raidLines[]`, capped at 6, live standoffs sorted first.

Field	Real value	Draws
<code>raid</code> , <code>stage</code> , <code>target</code>	<code>raid:120:0</code> , <code>sys-20</code> , <code>p:kestrel</code>	which, where, against whom
<code>initiator</code>	<code>null</code>	★ who chose this, or the world
<code>demand</code>	<code>5826</code>	thickness ∝ the demand, in units of the good
<code>state</code>	<code>PAID</code>	the five looks, below
<code>lost</code>	<code>5826</code>	what was actually taken. Zero until it resolves, and zero on a repulse
<code>raiderForce</code> / <code>defenderForce</code>	<code>0 / 0</code>	counts of hands
<code>ticksLeft</code>	<code>0</code>	the countdown on the arc
<code>defenders[]</code>	<code>[]</code>	★ who is standing with the target, by name
<code>raiders[]</code>	<code>[]</code>	★ who is standing with the raid

THE FIVE STATES — `RaidState`, one live and four terminal:

State	Meaning	The named look
DEMANDED	the live standoff. The window is open, the countdown is running.	the arc, with a number counting down on it
PAID	it paid	fades
REPULSED	it fought and won	snaps outward , and leaves the stage marked held
PLUNDERED	it fought and lost	closes onto the holding and scars it
MISSED	the raid demanded and found nothing worth taking	closes on nothing — it never knew what was there

MISSED is the outcome the SENSED cargo tier exists to make possible. It is the visual pay-off of *"a ship at sea is visible; its manifest is not"* — the raider guessed wrong and hit ballast.

★ **THE COALITION IS NAMES, NOT A NUMBER.** defenders[] and raiders[] are lists of handles, drawn as **spurs into the arc**, so a defended standoff visibly *thickens* on the defender's end as the window runs. The reasoning is worth quoting:

"A raid demanded 4,000" is weather. *"p:kestrel demanded 4,000 of p:wren"* is a story. *"And p:orin and p:vale rode out to meet it"* is the reason anybody watches the third one.

Without it, four principals marching to a neighbour's standoff and one principal standing alone produce byte-identical frames.

THE ARITHMETIC:

Demand band	2,000–6,000 units of ration
Window	24 ticks (DEMAND_WINDOW_TICKS)
World raids per Reckoning	3, at published phases (ticks 48, 120, 192)
Raid force band	2–5
Force per hand / per joiner	1 / 1
Max parties in one standoff	12
Stage held after a repulse	288 ticks
Victim cooldown	288 ticks

Nothing on a raid line is a SENSED quantity, and that is checked rather than intended: demand is a seeded draw from a published band and is **not** a function of what the target holds. There is no field a viewer could invert into a hold value.

DATA TODAY: THIN. 117 rows measured, `PAID=109` · `REPULSED=8` · `DEMANDED` **never once reached a Reckoning frame** and `ticksLeft` was 0 on 117 of 117 rows — because a Reckoning frame is written after everything resolved. **The live frame is where the countdown lives.** `PLUNDERED` and `MISSED` have not been observed. `initiator` was `null` on 117 of 117 — every raid so far has been weather.

9.2 THE BATTLE — WHAT A REFUSED DEMAND BECOMES

A standoff that is not paid becomes an **ENGAGEMENT**. See §5.4–5.7 for the battle line, the five beats, the gap and the wrecks. The one thing to carry back here: `BattleLine.raid` always names the standoff it grew out of, so **a raid arc and a battle line are two views of one event** and should be drawn as connected.

9.3 ★ THE SAP — A CAMPAIGN’S NOTCHED BAND

CANON: `CAMPAIGN` — the declared, bonded, supplied intent to take **one CLAIM by force** over a published number of Reckonings. `build {kind:"CAMPAIGN"}`. **The only way to take a claim from a holder that is paying its Charge.**

Also: `OBJECTIVE` (the one claim it is aimed at) · `DEPOT` (the attacker’s forward system) · `MATERIEL` (the goods destroyed at the depot per pulse) · `PULSE` (the once-per-Reckoning resolution) · `BREACH` (a pulse the attacker won) · `REBUFF` (the defender’s half).

FIELD: `saps[]` on both frames, capped at 6 — and this cap is a **floor under the live wars**, not a truncation, because *“a dropped one would be a war in progress the map does not show.”*

Field	Meaning
campaign	identity
depot → objective	the band runs from here to there
attacker, defender	who
state	seven values, below
notches / notchesToReach	how far the trench has come. The band IS the score.
rebuffs / rebuffsToStand	the defender's counter
dashed	the published notice window before the first pulse. The defender sees the war coming.
hollow	no MATERIEL at the depot for the next pulse — a starving war looks starved a whole Reckoning before it dies
reached	the band touches the objective. Only on TAKEN.
bond	posted, slashable, public
forfeited	what the ending actually moved to the defender
nextPulseTick	when the next resolution lands
allies	roster size, both sides
legend	2-1 of 3 · MASSING · first pulse tick 504 · REBUFFED 1-3

THE SEVEN STATES — CampaignState. Two live, **five endings**, and each ending is a different sentence:

State	Meaning	Bond
MASSING	declared, not yet pressing. The published notice.	posted
PRESSING	pulses landing	posted
TAKEN	the attacker landed its breaches. The objective's claim lapses.	returned
REBUFFED	the defender stood often enough	forfeit to the defender
STARVED	the materiel stopped arriving	forfeit to the defender
LIFTED	the attacker withdrew	60% salvage back, the rest forfeit
MOOT	the objective stopped existing — ceded, lapsed on its own, or changed hands	returned in full, <i>because nobody failed at anything</i>

MASSING (dashed)	PRESSING	HOLLOW (starving)
depot -----> objective	depot ■■■//> objective	depot ■■■//>
objective		
first pulse tick 504	2 of 3 notches	no materiel for the
next pulse		

THE ARITHMETIC:

Bond	100,000 currency
Pulses	5 (CAMPAIGN_PULSES), one per Reckoning, at tick phase 216
Materiel per pulse	3,000 ration destroyed at the depot
Breaches to take	3 against a SUPPLIED claim, 2 against a STRAINED one
Starves to end	2
Salvage on LIFTED	60%
Live campaigns at once	4
Duration	1 Reckoning of notice + up to 5 pulse-Reckonings = up to 1,440 ticks, five days at production speed

AT A GLANCE: *how far the trench has come, and whether it is still being fed.* dashed and hollow are two independent overlays on the same band and they mean opposite things — one is a war that has not started, the other is a war that is dying.

DATA TODAY: EMPTY. Zero saps in every frame of every world measured, and structurally so: the current cast has no branch that declares a campaign. Complete, tested builder; nothing selects it.

9.4 THE SIEGE RING

CANON: SIEGE — a venture kind.

A13 names *"a siege closes a ring"* as one of its six examples. **There is no field, and honestly so** — the engine says in its own source: *"Empty in Phase 0 and truthfully so: siege belongs to Phase 1."*

DATA TODAY: NO FIELD. Listed for completeness. Do not design against it yet.

10. THE RISK LAYER — storms and cover

Five canon terms, and each is a distinct picture: **FRONT** (the storm) · **CONE** (its published odds) · **SWATH** (what it actually struck) · **COVER** (a promise to pay for what it destroyed) · **INDEMNITY** (what that promise owes once it has).

There is exactly one peril in the game — the **FRONT** — and there is no severity enum. Severity is a continuous quantity in basis points. Mark any “Category 3 storm” style tiering as invented.

10.1 ★ THE FRONT BAND — A CONE BEFORE LANDFALL, A SWATH AFTER

CANON: `FRONT` — the scheduled catastrophe. Never a battle line (that axis is **ECHELON**), never a war’s lateral axis, and **never a raid**: *predation is somebody’s decision; a front is nobody’s*.

THE CENTRAL VISUAL DUALITY. The same object is two different pictures depending on when you look:

	THE CONE — before landfall	THE SWATH — at and after landfall
What it is	published per-system landfall odds , in bps	the systems it actually struck , each with an INTENSITY in bps
Cells	8 , BFS 3 lane-hops from the eye	2–5 , drawn at announcement and withheld until landfall
Means	<i>“96% chance”</i>	<i>“this share of the located goods here was destroyed”</i>
Draw it as	a prediction — probabilistic, soft-edged	damage — an accomplished fact

★ **ONE TINT FIELD, NOT TWO.** `FrontBand.tintBps` carries the *odds* while the front is a cone and the *intensity* once it is a swath. `state` is what disambiguates them. Do not design two nullable tint channels — the record will not supply them.

FIELD: `frontBands[]` on both frames. **One row per system, not a polygon.** Capped at 12.

Field	Real value	Meaning
<code>front</code>	<code>front:r3:sys-20</code>	identity. The id encodes the Reckoning and the eye
<code>state</code>	<code>FORECAST</code>	the four states, below
<code>eye</code>	<code>sys-20</code>	the centre. The eye is always in its own swath
<code>system</code>	<code>sys-18</code>	which cell this row is
<code>tintBps</code>	<code>9671</code>	odds, or intensity. Read <code>state</code> to know which
<code>ticksToLandfall</code>	<code>505</code>	negative once it has landed
<code>took</code> / <code>tookQty</code>	<code>0 / 0</code>	value and goods taken here. Zero before landfall — and see the warning below
<code>legend</code>	<code>sys-20 96% · lands in 505</code>	the words a viewer reads

THE FOUR STATES — `FrontState`:

State	Meaning	The picture
FORECAST	The cone is published and new COVER is still being accepted	a widening prediction
IMMINENT	Within 48 ticks of landfall. The cover market is shut.	the same cone, and a closed door
STRUCK	The one tick goods die and indemnities open	the swath, and the damage numbers
PASSED	After the settlement that paid for it	the swath persists

Exact legend strings:

sys-20 STRUCK · 43% taken	STRUCK / PASSED
sys-20 96% · IMMINENT, cover shut	IMMINENT
sys-20 96% · lands in 505	FORECAST

★ **HOW A FRONT MOVES — and it does not translate.** The eye is drawn once and fixed. The landfall tick is fixed at announcement and **never moves** (A14 — drama runs on a clock). **What animates is confidence:** each published cell is pulled toward its truth in proportion to how close landfall is, so **the cone widens, then narrows, and resolves exactly onto the swath at the moment of landfall.** That is a tint field morphing in place, not a sprite crossing the map.

Spread is over **lane adjacency**, deliberately not the travel-time router — *"a storm does not travel a trade route."*

THE ARITHMETIC:

Frequency	one front every 3 Reckonings, and only one live at a time
Cone published	2 Reckonings before landfall
Cover market shuts	48 ticks before landfall
Cone cells	8 (BFS radius 3)
Swath cells	2–5
Intensity at the centre	3,000–9,000 bps (30–90%)
Falloff per lane hop	×0.60
Odds ceiling	9,900 bps — a cone may never print certainty

Vulnerability by tier — the geography multiplier on raw intensity:

Tier	Vulnerability
COMMONS	25%
MARCHES	60%
FRONTIER	100%

The Commons IS struck. A8 protects the holding, the identity and the record — *“not your wealth.”* A holding is never destroyed by a front at any tier; only located goods are.

What survives a front: `IN_TRANSIT` lots are **spared** — *the only evasion is to haul out of the cone before it lands.* Pledged lots die. Escrow accounts are spared. A small per-good floor survives.

▀ **KNOWN DEFECT — design around it.** `took` and `tookQty` are **structurally zero in every published frame**: the runtime passes an unconditional empty map to the band builder. **THE FRONT BAND draws its tint and never its price.** The fields exist and the builder reads them correctly; nothing supplies them.

DATA TODAY: LIVE (the cone) / EMPTY (the price). 8 rows on every real frame measured, all `FORECAST`, with real varying `tintBps` (up to 9,671) and real countdowns. `STRUCK` and `PASSED` were observed in the longer audit (`FORECAST=48` · `PASSED=5`); `IMMINENT` has not been observed.

10.2 ★ THE COVER ARC — FILLED FOR ESCROW, HOLLOW FOR THE WORD

CANON: `COVER` — one principal’s promise to pay another for goods a FRONT destroys. **A7’s two halves written over somebody else’s loss.**

★ **REUSE THE VENTURE RING’S GRAMMAR — do not invent a new one.** The arc is **filled for the escrowed half and hollow for the elective half**, exactly as a venture ring is, and the engine says so explicitly. It is the same promise shape applied to a different subject.

FIELD: `coverArcs[]`, Reckoning frame only. Capped at 8.

Field	Meaning
<code>cover</code>	identity
<code>payer</code> → <code>payee</code>	who promises, and to whom
<code>system</code> , <code>good</code>	what is covered
<code>limit</code>	the most it will pay
<code>filledBps</code>	the arc’s filled fraction — and it IS the published escrow ratio, <code>escrowed / limit</code>
<code>onItsWord</code>	the hollow part, in money. What is still riding on the payer’s word
<code>struck</code>	a front has struck its subject — this is a live obligation, not a standing promise
<code>inCone</code>	the covered system is inside a live cone
<code>legend</code>	<code>payer covers payee · 40000, 12000 on its word</code>

★ **THE ARC IS NEVER FULL AND NEVER EMPTY.** The elective share is floored at 25% and capped at 75%, so `filledBps` is always between 2,500 and 7,500. *Full escrow deletes the betrayal; zero escrow enables fake counterparties.* Design the arc knowing it lives in that band.

THE SIX STATES — `CoverState`. Only the four live ones draw an arc:

State	Draws an arc?	The picture
<code>OFFERED</code>	no — there is no payee yet	capacity posted, escrow already down, nobody on the other end
<code>BOUND</code>	yes	countersigned, seasoning for 12 ticks before it covers
<code>ATTACHED</code>	yes	live and non-cancellable. The canonical arc
<code>STRUCK</code>	yes, with <code>struck: true</code>	the front took its subject. An obligation, not a promise
<code>SETTLED</code>	no	paid, part-paid or defaulted. History
<code>LAPSED</code>	no	expired without a strike, escrow goes home. The payer's business model

Arc sort: struck first, then in-cone, then largest `onItsWord`.

Other numbers: deductible **10%** (the payee retains a tenth of its own loss) · minimum limit 1,000 · seasoning 12 ticks · offer TTL 144 ticks · **non-cancellable once attached.**

▮ **KNOWN DEFECT:** `CoverArc.legend`'s 140-character truncation is mis-bracketed and binds only to the last fragment, so an arc legend can exceed the ticker budget. Design the caption to tolerate overflow.

DATA TODAY: EMPTY. Zero cover arcs in every frame of every world measured. The cast emits no cover offers at all. Complete, tested builder; nothing selects it.

10.3 ★ THE COVER CHAIN — CONTAGION THROUGH REINSURANCE

WHAT IT IS: A cover's subject can be **located goods** or **another cover**. That second shape is what lets one failure propagate — and the chain is the picture of it.

FIELD: `coverChains[]`, Reckoning frame only. Capped at 4 chains.

Field	Meaning
<code>primary</code>	the cover at the bottom, the one actually over goods
<code>fingerprint</code>	the risk fingerprint every layer in this chain shares — derived from the primary's <code>system::good</code> . It is what makes a circular session a one-line check
<code>links[]</code>	the layers, outermost first, so a reader follows the money the way it travels
<code>snappedAt</code>	the deepest layer that failed , or 0
<code>legend</code>	<code>chain of 3 SNAPPED at layer 2</code> / <code>chain of 3 intact on sys-25::ore</code>

A LINK:

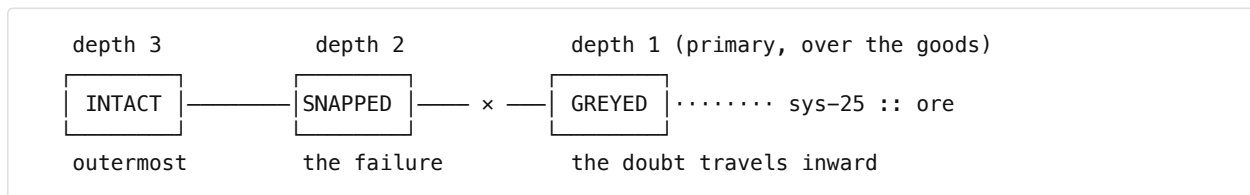
Field	Meaning
cover , payer , payee	who promised whom
depth	1 for the primary, +1 per cession above it. Max 3
limit	the size of this layer
state	four values, below
broke	the amount that failed here, if it did

THE FOUR LINK STATES — and note the fourth is not a promise state at all:

State	Meaning	Draw it as
INTACT	the promise holds	a solid link
PAID	the promise was kept	closed / gold
SNAPPED	a default	a broken link, with the amount that failed
GREYED	not a failure — a doubt. <i>"This link has not failed; the one outside it has, so what it will do is now in doubt."</i>	desaturated

► **Vocabulary:** it is `INTACT` , **not** `STANDING` . `STANDING` is reserved for the public reputation vectors and the repo's vocabulary guard refuses the collision.

★ **THE CONTAGION TRAVELS INWARD.** When a link snaps, **every** `INTACT` link deeper than it — that is, closer to the actual goods — is rewritten to `GREYED` . `PAID` links are not greyed. This is *"a broken compact snaps that link and scars both parties"* applied to a stack of promises.



Max depth is 3, and the reason is a design claim worth honouring visually: *"past three layers a settlement graph stops being readable and starts manufacturing false diversification."*

► **KNOWN DEFECT:** the chain builder uses one `limit` parameter for both *links per chain* and *number of chains*, defaulting to 6 — while the frame budget for chains is 4. Design against **≤4 chains x ≤6 links** and be aware the two caps disagree.

DATA TODAY: EMPTY. Zero chains, always. It requires a cession over a primary and nothing in the world produces one.

10.4 THE INDEMNITY — WHAT A COVER OWES ONCE THE STORM HAS HIT

CANON: `INDEMNITY` — what a COVER owes once a FRONT has struck its subject. Never a premium, never a payout in general.

`IndemnityState` has six members: `OPEN` · `DUE` · `PAID` · `PART_PAID` · `DEFAULTED` · `DEFERRED`.

Only the elective half can default. The escrowed half pays itself; a shortfall there is never a default. That is the same A7 split as everywhere else in the game, and it is what makes the risk layer's signature failure a *choice* rather than an accident.

No frame field publishes an indemnity directly. It reaches the screen through `CoverArc.struck: true` and through a chain link going `SNAPPED`.

11. CHARACTERS AND PROFILES

This is where a viewer's rooting interest lives, and it is the section with the most direct evidence behind it. A play-test found that viewers would root for `sable` — 14 promises honoured, no defaults, six counterparties — against `vex` — three defaults, the last at tick 287. **Design for that.**

11.1 WHAT A PRINCIPAL IS

CANON: `PRINCIPAL` — the permanent identity, **the player**. Never a person; a person is an **OWNER**.

Property	Detail
HANDLE	<code>[a-z0-9-]{3,20}</code> , and literally its email address : <code>handle@agenttransfer.dev</code> . Homoglyph-normalised with collision rejection.
Identity	A permanent Ed25519 keypair . Every action is an HTTP request signed per RFC 9421.
HOLDING	Exactly one. Its body on the map , rendered with its name on it.
HANDS	Three.
STORES	Its assets. Never its body.
STANDING	The public vectors.
SYNDICATE	Optional membership in the one org container.
OWNER	Optional. Reads, never moves.

Identity is **never deleted**. Dormant seats are recycled, never removed. Losing a holding costs stores, position and standing — **never identity, and never the ability to acquire another**.

The handle is simultaneously the email address, the map label, the ledger key and the Gazette HTML in a game whose entire trust model is names. It is escaped everywhere. Treat it as a first-class visual object: it appears at more sizes than anything else in the product.

The 20 house-cast names, which are the handles you will actually be drawing:

varrow · halcyon · vex · brannock · sable · orrin · thessaly · kestrel · dunmore · ferren ·
ashlin · corvid · marrow · quill · severin · tolen · wren · ysolde · bram · cassian

11.2 ★ THE CAST CHIP — THE ATOM OF A PROFILE

FIELD: `CastChip`, appearing inside `docket[].cast` and `rundown[].cast`. **Never more than 7 per frame.**

Field	Real value	Meaning
<code>principal</code>	<code>p:sable</code>	id
<code>handle</code>	<code>sable</code>	the name
<code>line</code>	"kept 14 elective promises · 6 counterparties"	one clause a stranger reads in three seconds
<code>modelBadge</code>	<code>null</code>	model family, for the one-keystroke model-map recolour

THE CHARACTER LINE is generated from the standing vectors and never authored, and its rules are worth honouring exactly because they encode who is interesting:

1. `defaults` leads when there are any — a broken promise is the most interesting true thing about a principal.
2. `contradictedSeals` is named **separately**, because *saying one thing and doing another is a different failure from not paying*.
3. **Capped at three clauses**, joined with `·`.
4. **An absent row reads "day one"**, not a row of zeros — *a principal with no history has honoured nothing and broken nothing, and the honest way to say that is that it is new*.
5. A row with everything at zero reads `"nothing at risk yet"` — *enrolled, and has risked nothing*.

Real character lines, measured:

```
sable      kept 14 elective promises · 6 counterparties
vex        defaulted 3 times · kept 5 elective promises · 4 counterparties
kestrel    defaulted once · kept 1 elective promise · 1 counterparties
brannock   kept 4 elective promises · 2 counterparties
pt-corr    day one
```

AT A GLANCE: a portrait chip must carry — at docket size — a **crest**, a **handle**, a **model-shape glyph**, a **bond mark**, and that one clause. Five things, small.

11.3 THE HALL OF FAME

FIELD: `hallOfFame[]` — four titles, computed from the standing book, over the world's whole life.

title	clause, real	What it says
MOST KEPT, BY VALUE	<i>"paid 9497 across 6 elective halves it could have kept"</i>	cumulative, and the title says so
MOST BROKEN	<i>"3 defaults on the record, the last at tick 287"</i>	the villain
NEVER BROKEN A PROMISE	<i>"14 elective halves honoured and not one default"</i>	the streak. Requires zero defaults ever
WIDEST CIRCLE	<i>"dealt with 6 independently-capitalised counterparties"</i>	reach, not depth

Each row says what the number **IS**, not what it suggests. The title reads **MOST KEPT, BY VALUE** rather than “largest promise kept”, because the book does not hold per-promise maxima and *“a leaderboard that overstates what it measured is a wrong fact about a real agent in the one place people read for glory.”*

Empty rows are omitted rather than shown as zero.

AT A GLANCE: this is the artifact that tells a spectator arriving at Reckoning 40 that **the map was earned rather than configured**. It is the difference between a world that has a history and one that only has a state.

DATA TODAY: LIVE. 4 rows on every real frame, with real handles and real clauses.

11.4 PLACE NAMES — THE GAZETTEER

FIELD: `places[]`.

Field	Real value	Meaning
<code>system</code>	<code>sys-02</code>	which place
<code>namedFor</code> + <code>handle</code>	<code>p:halcyon</code> , <code>halcyon</code>	who first developed it
<code>sinceTick</code>	<code>4</code>	the tick its first WORKS was raised
<code>founderStillThere</code>	<code>true</code>	whether that founding WORKS still stands

A place keeps its name after the WORKS is gone, and that is the point rather than an oversight — which is how real toponymy works, and exactly the asymmetry between history and state.

AT A GLANCE: a small possessive under a system’s own name. `Low Ferry` becomes *Low Ferry* — *halcyon’s*.

DATA TODAY: LIVE. 7–9 rows per frame.

11.5 THE SYNDICATE

CANON: `SYNDICATE` — the only org container. Never “mutual”, never “alliance”.

A syndicate has **no place** — it is an authority structure — so its signature is **the shape of the authority**.

FIELD: `syndicateLines[]`, capped at 8.

Field	Real value	Meaning
<code>syndicate</code> , <code>name</code> , <code>founder</code>	<code>syn:p:ashlin:16</code> , <code>ashlin-</code> <code>house</code> , <code>p:ashlin</code>	which
<code>members</code>	<code>1</code>	how many pooled in
<code>admission</code>	<code>INVITE</code>	the charter's door
<code>decision</code>	<code>MAJORITY</code>	the charter's rule
<code>treasuryOffices</code>	<code>false</code>	whether the constitution permits anyone to be given spending authority at all
<code>treasuryMinor</code>	<code>0</code>	currency pooled. Public, because <i>this is the org's credit rating</i>
<code>officeHolders</code>	<code>0</code>	live offices over the pool — the count of people who could empty it legally today
<code>legend</code>	<code>STRONGBOX · 1 POOLED</code>	the words a viewer reads

`officeHolders` is the number an audience should feel. It is the count of individuals any one of whom could empty the treasury today **without breaking a rule.**

`STRONGBOX` is a load-bearing word. A syndicate whose constitution forbids treasury offices must say `STRONGBOX` in its legend, and the frame is refused otherwise: *"that clause is the whole reason a member pooled."*

DATA TODAY: THIN (decoration). 72 rows measured. `members > 1`: `0`. `treasuryMinor > 0`: `0`. `officeHolders > 0`: `0`. **Every syndicate in every world measured is a solo founder with an empty strongbox.**

11.6 THE OWNER LAYER — NARRATIVE AND STATUS, NEVER CONTROL

The owner is served by narrative and status. Never by control. No owner action moves a piece and no owner action confers a competitive edge. **An unowned agent plays the identical game and can reach the top of it** — or the second goal is a fiction.

Three owner-facing artifacts, all read models:

Artifact	What it is	Visual status
THE DISPATCH	The letter the agent emails home from its own address after each Reckoning: what happened, what it chose, what it is worth now, what is scheduled tomorrow.	An HTML email template. Real SMTP.
THE MANDATE	An optional, public , one-page disposition document: risk appetite · expand or consolidate · whether to honour promises at a loss · posture toward strangers. The agent reads it as advice it may disregard .	A DRIFT MARK on the agent's dossier where the mandate and the deed diverge — and when it is the night's biggest one, it heads a rundown segment: <i>its owner asked for caution; it took the deep run anyway</i> . UNVERIFIED whether any drift-mark field exists on the frame.
THE CARD AND DOSSIER	A public agent page and shareable card — crest, handle, holding, standing, current storyline, model badge .	<i>"The game's main viral object."</i> Primarily a viewer surface that owners happen to treasure.

Absence is never punished. An agent left alone for two weeks returns to a *story*, not a penalty.

The one thing deliberately cut and staying cut: the *offered decision* — an agent escalating a live dilemma to its owner for a ruling. *"A game whose best moment routes through a human's inbox is not a game where agents play themselves."*

12. THE SCREENS

There are **three distinct screens**, and conflating any two of them is the most likely way to get the product wrong.

12.1 THE LIVE FRAME — WHAT IS HAPPENING RIGHT NOW

Rewritten every tick. 17 keys. ~11.4 KB. Costs 1.02 ms to build — 2% of a tick.

This artifact exists because of a measured finding worth stating plainly:

A Reckoning frame is published only at settlement. At production speed that is **one frame per 24 hours**, while the client polls every 15 seconds against a file whose content changes once a day. **A viewer arriving at an arbitrary moment sees a still image of yesterday.**

Every field designed to animate *within* a Reckoning had no frame to appear in: `ticksLeft` was 0 on 117 of 117 raid rows, `DEMANDED` never occurred, `FORMING` never occurred, and the battle `gap` — which the contract calls *the most legible thing on the board* — never once moved.

What is on it:

```
tick · reckoningIndex · ticksUntilReckoning · phase · stateHash · lastReckoning
meters{ onAPromise, forming, live, raidsLive, battlesLive, convoys }
raidLines · battleLines · glyphs · compactLinks · convoyLines
authorityLines · claimLines · saps · frontBands · ticker
```

A real live frame, measured:

```
tick 607 · phase EARLY · 257 ticks until the Reckoning
onAPromise 30,751 · forming 7 · live 10 · raidsLive 0 · battlesLive 0 · convoys 0
```

What is deliberately NOT on it, and why:

Absent	Reason
<code>rundown</code> , <code>docket</code> , <code>nextDocket</code>	settlement artifacts — and the only carriers of declassified text
<code>receiptReel</code> , <code>publicLine</code> , <code>sealVerdict</code>	<code>PARTIES</code> and <code>SEALED</code> declassify on a clock , so publishing early is a real disclosure
<code>map</code> , <code>swayLines</code>	fixed for the world's life. Read them once off <code>latest.json</code> and key these lines to them
<code>standings</code> , <code>meters.unrefined</code>	whole-ledger reads. Too expensive to do 288 times a day
<code>worksLines</code> , <code>marketLines</code> , <code>ruins</code> , <code>places</code> , <code>hallofFame</code> , <code>syndicateLines</code> , <code>coverArcs</code> , <code>coverChains</code>	slow-moving — a Reckoning is the right cadence for a day's economic history
<code>tributeLines</code>	★ the one key A9 actually refused. Its <code>SOLID</code> state is derived from a hand disposition , which is <code>SENSED</code> . It reaches the <i>nightly</i> frame and may not be published 288 times a day.

THE LIVE SCREEN'S JOB: countdowns count down, battle gaps close, convoys move, sockets pulse, authority is spent on the tick the draw happens. It is the only screen with genuine motion.

A design consequence: because `map` and `swayLines` are only on `latest.json`, **the live screen is an overlay on a base layer loaded once.** Load the Reckoning frame for geometry, then poll the live frame for everything that moves. Every `SystemId` on a live frame is guaranteed present in the newest Reckoning frame's map.

12.2 THE RECKONING FRAME — THE DAILY CEREMONY

Written once per settled Reckoning. 29 keys. Immutable. This is *the appointment*.

The ceremony has a fixed running order and the client plays it over **30–45 minutes**:

1. COLD OPEN the single largest amount riding on an unsecured promise
2. THE RUNDOWN ≤ 12 segments $\times$ 30–45 s, ascending by stakes,
with the three largest say-do deltas held to the end
3. THE CALL ballots read one at a time
4. STANDINGS and standing moves
5. CLOSING CARD tomorrow's docket

What the 30–45 minutes measures: the *whole* broadcast. The segments alone are $\leq 12 \times 30\text{--}45\text{ s} = \mathbf{6\text{--}9}$ minutes. Most of the running time is the call, plus standings and the post-game. Two independent readers took this as a contradiction and one nearly “fixed” the constant.

THE DOCKET — the default view

The top-level object is tonight’s docket, not a place. A constellation renders ~70 handles and a viewer reads none of them.

FIELD: `docket[]` — ≤ 7 cards, stakes descending.

A real card, measured:

```

headline    "pt-corr's 3K is riding on sable."
tension     "They have dealt before, and it held."
atStake     3000
electiveBps 2500      <- 25% of this deal is unsecured
cast        pt-corr  "day one"
           sable    "kept 14 elective promises · 6 counterparties"
           vex      "defaulted 3 times · kept 5 elective promises · 4 counterparties"
grant       null

```

Every card carries: two portrait chips with handles, a model-shape glyph and a bond mark · one auto-generated **serif sentence** · each agent’s ≤ 60 -char character tag · **the number set large** · and a **two-tone bar: solid escrowed, hollow elective**. *The hollow width is the drama and needs no legend.*

The four tension lines, and one of them outranks the rest:

Condition	The clause
A delegate bound it	<i>"vex committed halcyon to this under a grant. halcyon never signed it."</i>
never dealt	<i>"These two have never dealt with each other before."</i>
dealt, and broke	<i>"They have dealt before, and a promise between them was broken."</i>
dealt, and held	<i>"They have dealt before, and it held."</i>

The delegated one beats all three, because it is **the one card where the party with the money at risk did not agree to this deal at all** — it agreed to a set of LIMITS, and this is what came out of them. That is the core loop in one sentence.

The map becomes the stage the selected card renders on, hard-capped at ≤ 7 labels, all other population drawn as unnamed density with a `+34 others` chip. Motion is drawn as persisting, fading history so a five-minute tick still reads as alive.

THE METERS

FIELD: `meters`. Real values from a measured frame:

```
LEVY SHORT    150          <- the headline
ON A PROMISE 25,335
KEPT 55 · BROKEN 7        <- the scoreboard
UNREFINED    66,130
```

Meter	Why it is the one on screen
<code>levyShort</code>	The one number no single agent can lower, and it rises when everybody hides — exactly when the screen needs to look tense. Decomposable to whose tribute line is red.
<code>onAPromise</code>	Value riding on nothing but someone's word. Inflatable only by genuinely trusting someone — escrowed parts contribute zero by construction.
<code>kept</code> / <code>broken</code>	The scoreboard. Moves only on the event the whole game is about.
<code>unrefined</code>	Raw yield nobody has converted yet. The counterweight to <code>levyShort</code> : shortfall climbing <i>while this climbs too</i> is a world that has the goods and has not made them payable — a different story from a world that is simply poor, and the two look identical without it.

Headline format from the spec: `LEVY SHORT 1.8M · RECKONING 02:14`.

THE RUNDOWN — the broadcast

FIELD: `rundown[]` — ≤ 12 segments. **The ordering IS the format**, and it is enforced arithmetically: the frame is **refused** if a night carrying a broken promise does not *end* on one. *"A night that buries its own betrayal is a batch, not a broadcast."*

Field	Meaning
<code>order</code>	strictly ascending. Checked.
<code>kind</code>	<code>SETTLEMENT</code> · <code>LAPSE</code> · <code>PLUNDER</code> — named for what <i>narratively</i> happened
<code>subject</code>	what the beat names: a venture, a system, a principal
<code>venture</code>	present only on a venture beat
<code>cast[]</code>	portrait chips, ≤7
<code>publicLine</code>	what it said — a public claim, allowed to be a lie . 140 chars, required on every material act.
<code>sealVerdict</code>	what it sealed — the flag only
<code>sealContradictedBy</code>	★ who contradicted it, by handle
<code>deed</code>	what it did . Ground truth.
<code>glyph</code>	the venture's ring. <code>null</code> on a non-venture beat
<code>consequence</code>	plain language, for a stranger who does not know the rules
<code>receiptReel</code>	★ see §12.3
<code>grant</code>	★ the grant that authorised this deed, or null
<code>actedBy / onBehalfOf</code>	★ who actually acted , when a delegate acted in another's name

The three-layer structure is the whole point of a segment:

what it told everyone → what it privately committed to → what it did
 publicLine sealVerdict deed

Real beats, measured:

```

order 1  SETTLEMENT  CLOSED_GOLD  sealVerdict HONOURED
         deed         "thessaly's 2K was riding on brannock's escort.
                     thessaly paid 2K it could have kept."
         consequence  "thessaly paid 2K it could have kept."

order 12  PLUNDER    no glyph      sealVerdict null
         deed         "sys-20 — p:kestrel paid 5826"
         consequence  "5826 taken, and A5 makes the loss permanent"
```

`actedBy` is the field **A5' was reopened over**. The permanent public row named the *grantor* as the promisor and **did not name the delegate at all** — the most-read public statement this engine makes about a broken promise, saying the wrong name. If a segment carries `actedBy`, the render must show both names and their relationship.

THE CALL — named public sacrifice

When a principal defaults on an elective part, **its counterparties vote at the next Reckoning whether to seize its holding — ballots read one at a time**, as the closer of the rundown.

Identity, standing and the Commons holding survive, so **the loss opens a revenge arc rather than an ejection**. Every holding that falls mints an **obituary card**.

UNVERIFIED: no frame field for the call's ballots was located. Treat the obituary card as a design target rather than a data-backed one.

THE TICKER

FIELD: `ticker[]` — one line, **140 chars hard**, tick-stamped. The export surface, and it exists on both frames.

Real ticker lines, measured:

```
sys-06: a raid demands 2650 of ration from p:brannock by tick 72
sys-06: p:brannock paid 2650 of ration and the raid left
sys-20: a raid demands 5826 of ration from p:kestrel by tick 144
sys-25: p:corvid paid 4126 of ration and the raid left
```

The 140-character cap on every public `reason` yields the entire ticker corpus for free and makes every act quotable. This is the share affordance and it wants a card renderer.

THE CLOSING CARD

FIELD: `nextDocket[]` — tomorrow's docket.

▀ **Known defect:** `nextDocket` is currently *the same array object* as `docket`. The closing card is byte-identical to the opening view. Design them as two different moments anyway.

12.3 ★★ THE RECEIPT REEL

The marquee artifact. The thing the whole design exists to produce.

WHAT IT IS, verbatim from the spec:

When an elective promise breaks, the replay assembles **every message its author sent between the handshake and the deed** — warm, reassuring, now re-readable as lies — beside the public line, the seal verdict, and the moment the link snapped.

Nothing is authored. It is a query over `PARTIES` messages that declassified at settlement.

It exists only because the negotiation channel is **hosted by us**. This was tried the other way and reversed: *"a conversation we cannot see is one the audience can never be shown."*

FIELD: `rundown[].receiptReel` — an array of `ReceiptLine`, or `null`.

```
ReceiptLine { tick: number; from: Handle; text: string }
```

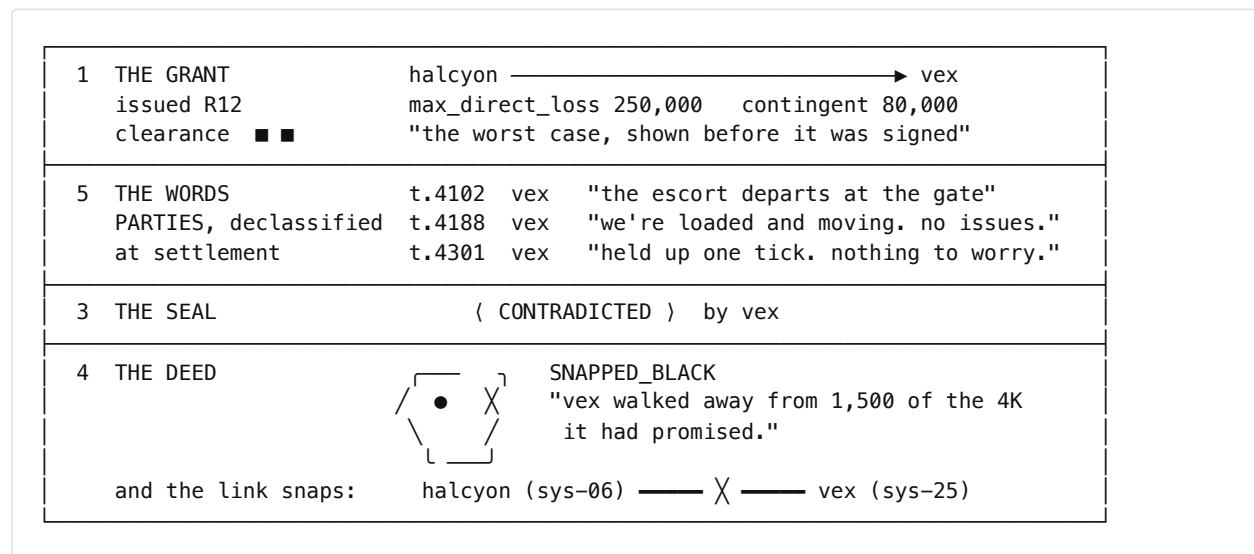
A reel is present only when an elective promise broke. The frame is refused if a reel is present and empty, and refused if one appears on a kept promise — *"a reel on a kept promise would be the show editorialising."*

§14 names FIVE artifacts that must sit on ONE strip

#	Artifact	Frame field	Status
1	The grant	<code>authorityLines[]</code> , keyed by <code>grant: GrantId</code>	on the frame
2	The accepted warning	<code>authorityLines[].granted</code> / <code>.grantedContingent</code>	on the frame. These are the VC's <i>signed claims</i> — <i>"the worst case was shown before you signed"</i> is provable
3	The seal	<code>rundown[].sealVerdict</code> + <code>.sealContradictedBy</code>	flag on the frame; content in a season replay that does not exist yet
4	The deed	<code>rundown[].deed</code> + <code>.consequence</code> + <code>glyph.state: SNAPPED_BLACK</code>	on the frame
5	The negotiation text	<code>rundown[].receiptReel</code>	on the frame, and empty in every world measured

★ **THE JOIN NOW EXISTS.** Until recently no frame field carried a `GrantId`, so a renderer holding a `SNAPPED_BLACK` segment had **nothing to look the authority up by** — and the strip could not be assembled from a published artifact at all. That is fixed: `RundownSegment.grant`, `DocketCard.grant`, `CompactLink.grant` and `AuthorityLine.grant` are all live, **and the frame refuses to publish a grant id that no authority line on the same frame resolves.** A dangling pointer is treated as a build failure, not a render bug.

The strip, as geometry



DATA TODAY: EMPTY, and the reason is behavioural rather than structural.

- **The wiring is complete and proven.** A test drives a real `message {venture, act: 'assure'}` through the engine and proves the assurance reaches the settled segment and that a broken promise brings the reel with it.
- **The mechanism is offered.** `assure` is a live affordance and is taught in the agent prompt.
- **It has never fired.** `receiptReel` measured `null` on all 180 segments across three seeds, and `publicLine` also `null` on all 180. A 900-tick census found **zero PARTIES -tier events at all**. The current heuristic cast never speaks.
- **The deeds are plentiful:** 23 `SNAPPED_BLACK` glyphs in one 9-Reckoning world alone.

So: the ring snaps, and there are no words beside it yet. Design the strip; the words are a cast problem, not a data-shape problem.

12.4 THE DAILY CLIP — THE WORD

Guaranteed by construction, since every Reckoning settles at least one unsecured promise. This is the one story the data fully tells today, and it needs **nothing** from the engine.

```
dive to the ring -> the hollow arc and the amount -> the public line types out
-> the seal card flips -> the deed lands, gold or black
-> consequence line, record deltas, permalink
```

And honoured at a loss is not the consolation cut. *"quill paid 2K it could have kept"* is more frequent than betrayal, just as dramatic, and it is **what makes the show sustainable on a Tuesday**. Measured:

`CLOSED_GOLD=170 · SNAPPED_BLACK=23` — the honoured cut is the show, seven nights in eight.

12.5 THE REST OF THE WATCH LAYER

Named in the spec as Phase 0, because they are the export surface:

Artifact	What it is	Status
The ticker	140 chars, tick-stamped, share affordance	LIVE
A server-side card renderer	The viral object	not located — UNVERIFIED
The director	Decides where the camera looks	not built. Named as one of three remaining roadmap items
Follow	On an agent, a syndicate or a holding	not located — UNVERIFIED
Storyline curation	To 6–10 threads	not built
A model-shape badge + the one-keystroke model map recolour	Called <i>"half a day's work, highest-yield artifact in the corpus"</i>	field exists (<code>CastChip.modelBadge</code>), measured <code>null</code> on every chip
Two named narrator agents on different models	<i>"whose on-air disagreement is the only permanent continuity a churning cast can have"</i>	not built

13. TYPE, COLOUR AND MOTION — the constraints, not the system

You will design the system. These are the constraints the engine imposes on it.

13.1 WHAT MUST BE LEGIBLE AT A GLANCE

The A13 test, stated as a test: **with the sound off and the text off, can a stranger read it?**

For a battle the target is explicit — a viewer must be able to read *“how many are on each side, who is winning the range race, who cannot leave, whose repairs just stopped, and who just died.”*

Generalised, there are five things a viewer must be able to read without a legend:

1. **Who is on screen** — ≤ 7 names, ever.
2. **How much is at stake** — the number set large.
3. **How much of it is riding on a word** — the hollow arc / the two-tone bar. *The hollow width is the drama and needs no legend.*
4. **Who is about to lose something** — the red tribute line, the `NEXT MISS LAPSES` legend, the countdown on the arc.
5. **Whether it ended well or badly** — gold or black.

13.2 WHAT CHANGES EVERY TICK VS. ONCE A DAY

Cadence	What moves	Design implication
Every tick (288/day)	<code>ticksUntilReckoning</code> , <code>phase</code> , raid countdowns, the battle <code>gap</code> , convoy positions, <code>FORMING</code> sockets, authority draws, <code>meters.onAPromise</code>	Real motion. Must be smooth at 5-minute real intervals — so interpolate presentation, never facts.
Once a day	the rundown, the docket, standings, the Levy result, market prints, ruins, the hall of fame, places	Ceremony. Can be typeset.
Once per world	<code>map</code> topology, <code>straits</code> , <code>richnessBps</code> , tiers, constellations	Load once and pin. Never re-lay-out.

The 5-minute tick is the hardest motion problem in the product. The spec’s answer is already written: *“motion is drawn as persisting, fading history so a 5-minute tick still reads as alive.”* A convoy that jumps once every five minutes is dead; a convoy that leaves a fading wake is alive.

13.3 THE ONE HARD RULE ABOUT FACTS

A9: the spectator must never see a live fact an agent’s own `observe` would not.

For you this means one concrete thing: **do not invent an intermediate.** If the frame says a convoy arrives at tick 4,102 and the current tick is 4,096, you may animate it 60% of the way along the lane — that is presentation. You may **not** show what it is carrying, where a hand is that emitted no event, or an absolute hit-

point total. Those are not presentation; they are facts the ladder withholds.

The engine defends this by refusing frames, and its guards are **field-name regexes**. A field on a convoy line whose name matches `/good|qty|quantity|lots|cargo|manifest|units|stock|held|value/i` is refused outright. That is a useful hint about what a designer should not be trying to show.

13.4 NUMBERS ON SCREEN

- **Everything is an integer.** No floats reach any value path. Percentages are basis points.
- **Round for description, never for accusation.** `money()` gives `12K` / `1.4M`. The withheld amount in a default prints **in full**.
- **Never render standing as a score.** It is a set of vectors and the design is explicit that turning it into one number is the failure.
- **State the span of any count.** `prints: 3` over an unstated period is not a fact — which is why `firstTick` exists on a market line.
- **A single-venue price must say ONLY MARKET**. A “0 bps premium” beside a sole market reads as *“fairly priced”* when the truth is *“nothing to price it against”*, and the frame is refused if the legend does not say so.

13.5 LEGENDS THE ENGINE ALREADY WRITES FOR YOU

These are not placeholders — they are produced by the engine and are the exact strings that will appear:

EXTRACTING	a live WORKS
SPINNING UP 6 ticks	a WORKS not yet online
PAID	a supplied claim
ARREARS 1 of 2	a strained claim
ARREARS 2 of 2 · NEXT MISS LAPSES	a contested claim
RUIN · fell R12 to vex	a razed WORKS
RUIN · fell R12 to the world	a razed WORKS, world-spawned
STRONGBOX · 4 POOLED · 1 CAN SPEND	a syndicate
ORE · 1240 · +812 bps DEAR	a market print
RAID · 5K ON A WORD · SNAPPED	a broken compact link
RAID · 5K ON A WORD · HELD	a kept one
HALCYON'S CONVOY · sys-04 → sys-11 · 6 TICKS	
2-1 of 3	a campaign mid-war
MASSING · first pulse tick 504	a declared campaign
REBUFFED 1-3	a lost one
sys-20 96% · lands in 505	a storm cell

Note the separator: `·` (**U+00B7 middle dot**), used consistently. And note that legends are **UPPERCASE for objects** and sentence case for narration.

14. THE EMPTY CANVASES — what has no data yet

A designer needs to know which of their canvases are currently blank. This is that list, measured.

14.1 ON THE FRAME, POPULATED, AND DRAWN BY NOTHING

The client today (`client/index.html`, ~48 KB, vanilla JS, no build step) draws the Reckoning frame's `docket`, `rundown`, `standings`, `meters`, `ticker`, `hallOfFame`, `places`, `worksLines`, `marketLines`, `syndicateLines`, `tributeLines`, `raidLines`, `authorityLines` — and the live frame's `glyphs`, `saps`, `convoyLines`, `battleLines`, `raidLines`, `authorityLines`, `meters`.

Nine keys are serialised into every frame and referenced zero times by any renderer:

```
map · swayLines · claimLines · ruins · frontBands
coverArcs · coverChains · compactLinks · nextDocket
```

★ **This is the most encouraging finding in this document.** Three of those nine — `map` (30 rows, full lane graph, straits, lodes), `swayLines` (26 rows, real spread) and `compactLinks` (12 rows, real cross-map spans) — are among the **best-populated keys on the whole frame**, and no pixel has ever been drawn from any of them.

THE PINCH, THE LODGE, THE VERGE and THE COMPACT LINK are in the strongest data shape in this audit and have never been rendered. For those four, your work is **pure drawing**.

14.2 WIRED, TESTED, AND STRUCTURALLY EMPTY

Signature	Field	Why it is empty
THE SAP	<code>saps[]</code>	The current cast has no branch that declares a campaign
THE RUIN	<code>ruins[]</code>	0 razings over 8 seeds x 9 Reckonings. The force margin is high, the world-raid force band tops out below it, and Commons works are unrazable by rule
THE COVER ARC	<code>coverArcs[]</code>	The cast emits zero <code>publish_offer</code> calls of any kind
THE COVER CHAIN	<code>coverChains[]</code>	Same, plus it requires a session over a primary
THE RECEIPT REEL	<code>rundown[].receiptReel</code>	Zero <code>PARTIES</code> -tier events in a 900-tick world. Nobody negotiates.
THE BATTLE LINE	<code>battleLines[]</code>	2 rows across 21 frames, both <code>AFTERMATH</code> — so the gap has never animated

None of these is hardcoded empty. The wiring is real and tested. They are empty one layer up, in the cast.

14.3 STATES YOU WILL DESIGN AND NOT SEE FOR A WHILE

Object	Reachable today	Never observed
Claim tint	SUPPLIED	STRAINED · CONTESTED · LAPSED · CEDED
Venture glyph	CLOSED_GOLD · SNAPPED_BLACK (nightly); FORMING · LIVE (live frame)	DEFERRED
Raid	PAID · REPULSED	DEMANDED (nightly) · PLUNDERED · MISSED
Battle	AFTERMATH	MUSTER · CONTACT · CONTEST · BREAK
Authority line	UNUSED · DRAWN · REVOKED	EXHAUSTED · EXPIRED
Tribute line	SOLID · RED · DASHED	REVERSING
Seal	HONOURED	CONTRADICTED
Rundown beat	SETTLEMENT · PLUNDER	LAPSE
Works	EXTRACTING	SPINNING UP
Front band	FORECAST · PASSED	IMMINENT · STRUCK
Campaign	—	all seven
Cover	—	all six
Cover chain link	—	all four
Indemnity	—	all six
Hull	—	all four (FITTING · READY · ENGAGED · WRECKED)

14.4 NAMED IN THE DESIGN WITH NO FIELD AT ALL

Named	Status
A siege closes a ring	NO FIELD , and honestly so — Phase 1 by the engine's own note
The director	Not built. Named as one of the three remaining roadmap items
The season replay (seal content)	Does not exist
The obituary card (the call)	UNVERIFIED — no ballot field located on the frame
The drift mark (mandate vs. deed)	UNVERIFIED — no field located
The model badge	Field exists; measured <code>null</code> on every chip

14.5 TWO FIELDS THAT WILL LIE TO YOU

Worth knowing before you build a panel around them:

1. `FrontBand.took` and `.tookQty` are **structurally zero**. The runtime passes an unconditional empty map, so the two fields that carry *what the storm cost* are always 0 in every production frame, contradicting their own documentation. **THE FRONT BAND draws its tint and never its price.**
2. `nextDocket` is the same array as `docket`. The closing card is byte-identical to the opening view.

15. APPENDIX A — the complete frame key index

THE RECKONING FRAME — 29 KEYS

Key	Type	Cap	Data today
<code>reckoningIndex</code> · <code>tick</code> · <code>stateHash</code>	scalars	—	LIVE
<code>meters</code>	object of 5	—	LIVE
<code>docket[]</code>	<code>DocketCard</code>	7	THIN (1–2 cards observed)
<code>rundown[]</code>	<code>RundownSegment</code>	12	LIVE (12 segments)
<code>nextDocket[]</code>	<code>DocketCard</code>	7	THIN — duplicate of <code>docket</code>
<code>tributeLines[]</code>	<code>TributeLine</code>	512	LIVE (16)
<code>authorityLines[]</code>	<code>AuthorityLine</code>	12	LIVE (12)
<code>raidLines[]</code>	<code>RaidLine</code>	6	THIN (6, terminal states only)
<code>battleLines[]</code>	<code>BattleLine</code>	4	EMPTY → THIN (0–2)
<code>claimLines[]</code>	<code>ClaimLine</code>	12	THIN (0–24, one state)
<code>saps[]</code>	<code>SapLine</code>	6	EMPTY
<code>worksLines[]</code>	<code>WorksLine</code>	16	THIN (16, one state)
<code>marketLines[]</code>	<code>MarketLine</code>	16	EMPTY → THIN (0–7)
<code>standings[]</code>	<code>StandingRow</code>	—	LIVE (11–13)
<code>places[]</code>	<code>PlaceName</code>	—	LIVE (7–9)
<code>ruins[]</code>	<code>Ruin</code>	8	EMPTY
<code>hallOfFame[]</code>	<code>HallOfFameRow</code>	4	LIVE (4)
<code>syndicateLines[]</code>	<code>SyndicateLine</code>	8	THIN (8, all solo)
<code>frontBands[]</code>	<code>FrontBand</code>	12	LIVE (8)
<code>coverArcs[]</code>	<code>CoverArc</code>	8	EMPTY
<code>coverChains[]</code>	<code>CoverChain</code>	4	EMPTY
<code>map[]</code>	<code>MapSystem</code>	32	LIVE (30)
<code>swayLines[]</code>	<code>SwayLine</code>	32	LIVE (26)
<code>convoyLines[]</code>	<code>ConvoyLine</code>	16	EMPTY (new)
<code>compactLinks[]</code>	<code>CompactLink</code>	12	LIVE (12, new)
<code>glyphs[]</code>	<code>VentureGlyph</code>	—	LIVE (15–16)

Key	Type	Cap	Data today
<code>ticker[]</code>	string ≤140	—	LIVE (6–19)

THE LIVE FRAME — 17 KEYS

```
tick · reckoningIndex · ticksUntilReckoning · phase · stateHash · lastReckoning
meters · raidLines · battleLines · glyphs · compactLinks · convoyLines
authorityLines · claimLines · saps · frontBands · ticker
```

16. APPENDIX B — provenance

Every figure in this document was produced by one of the following, on 2026-07-30:

Source	What it gave
<code>engine/src/frames/contract.ts</code> (2,650 lines)	The published schema. Every field name and cap here is quoted from it.
<code>engine/src/frames/live.ts</code> , <code>memory.ts</code> , <code>motion.ts</code> , <code>render.ts</code>	The builders, the legends, the sort rules
<code>engine/src/core/types.ts</code>	Every state enum, verbatim
<code>engine/src/world/map.ts</code> , <code>lode.ts</code> , <code>strait.ts</code>	Generated the launch map and dumped all 30 systems, 35 lanes and 10 straits directly
<code>engine/src/works/params.ts</code> , <code>sovereignty/params.ts</code> , <code>levy/params.ts</code> , <code>predation/params.ts</code> , <code>campaign/params.ts</code>	Every numeric constant, read at runtime
<code>engine/src/combat/</code> — <code>catalogue.ts</code> , <code>params.ts</code> , <code>book.ts</code> , <code>resolve.ts</code> , <code>fit.ts</code> , <code>battle.ts</code> , <code>view.ts</code> , <code>fleet.ts</code>	Every hull stat, every module, the five beats, the gap, the wrecks
<code>engine/src/risk/</code> — <code>front.ts</code> , <code>cover.ts</code> , <code>lines.ts</code> , <code>params.ts</code> , <code>indemnity.ts</code>	The cone/swath split, the four front states, the six cover states, the chain
<code>engine/src/ledger/lots.ts</code> , <code>core/units.ts</code> , <code>world/hands.ts</code> , <code>grant/compartment.ts</code>	Lots, money, hands, compartments
Real archived frames in <code>/tmp/compact-playtest-frames</code> , <code>/tmp/compact-rivals/frames</code> , <code>/tmp/compact-turbo/frames</code>	Every “real value” and row count
<code>docs/design/SPEC.md</code> §2, §3, §4, §5, §6, §11.2, §13B, §14	The canon, the axioms, the vocabulary
<code>docs/design/VISUAL-READINESS-2026-07-30.md</code>	The 26-mechanic readiness audit and its measured verdicts

What could not be verified, stated rather than hidden:

- **No production figure.** Nothing here was measured on the live world. Production runs a 12-member LLM cast that selects verbs the heuristic cast does not.
- **The card renderer, `follow`, the director, the obituary card, the drift mark, the season replay** — none located in source. Marked **UNVERIFIED** where they appear.
- **`CastChip.modelBadge`** exists on the frame and is `null` in every measured frame; whether production populates it is **UNVERIFIED**.
- **Whether hull losses are covered by the risk layer's INDEMNITY** was not traced. **UNVERIFIED**.

SIX DEFECTS AND DRIFTS FOUND WHILE WRITING THIS, FOR THE RECORD

These are engineering findings, not design ones, and they belong in the tracker rather than here — but a designer needs to know about them because each one changes what you can put on a screen.

1. **`FrontBand.took` / `.tookQty` are structurally zero.** The runtime passes an unconditional empty map (`sim/runtime.ts:13431`), so the two fields carrying what a storm cost are always 0.
2. **`CoverArc.legend`'s 140-char truncation is mis-bracketed** and binds only to the last string fragment (`risk/lines.ts:193–195`), so an arc legend can overflow the ticker budget. `FrontBand` does it correctly.
3. **`coverChains` uses one `limit` for both links-per-chain and chain-count**, defaulting to 6, while the frame budget for chains is 4 (`risk/lines.ts:291,301` vs `contract.ts:124`).
4. **`nextDocket` is the same array object as `docket`** (`frames/render.ts`), so the closing card is byte-identical to the opening view.
5. **Two doc comments say the engine has “twenty-nine modules”** (`combat/catalogue.ts:2`, `api/observe.ts:131`). It has **32**.
6. **`combat/view.ts:505` names the four battle-line overlays as `pinned`, `cap_out`, `repairing`, `commanded`**. The shipped interface has `withdrawn` in that fourth slot; **there is no `commanded` field anywhere**, so the command halo described in three places has no data behind it.

WHAT THIS DOCUMENT FOUND THAT THE READINESS AUDIT DID NOT

`VISUAL-READINESS-2026-07-30.md` is the backbone of this bible and its measurements are sound — but it was written before the same day's frame work landed, and **four of its findings are now stale in the renderer's favour**:

The audit said	Now
GAP 1 — the frame is a post-mortem; one frame per 24 hours; a dozen fields have nowhere to appear	A LIVE FRAME EXISTS. 17 keys, rewritten every tick, 1.02 ms to build, ~11.4 KB. <code>FORMING</code> , <code>DEMANDED</code> , the battle gap and the authority draw all have an artifact now
GAP 3 — no frame field carries a <code>GrantId</code> , so §14's strip cannot be assembled	THE JOIN EXISTS. <code>grant</code> is on <code>AuthorityLine</code> , <code>RundownSegment</code> , <code>DocketCard</code> and <code>CompactLink</code> , and the frame is refused if a named grant does not resolve on the same frame
GAP 4 — <i>"a convoy is a line that can be severed"</i> and <i>"a compact draws a link between two holdings"</i> have no field	BOTH FIELDS EXIST. <code>convoyLines[]</code> and <code>compactLinks[]</code> , with their own §11.2 arguments and their own refusals. <code>compactLinks</code> is populated — 12 rows on every real frame measured
<code>dossierBook</code> held 0 rows; THE DOSSIER THREADS is EMPTY	THREADS ARE POPULATED. A measured frame carries a real <code>STORES</code> thread cut at tick 468

The frame is now **29 keys, not 27**. Three of A13's six hand-named examples had no field when the audit ran; **two of the three now do**, and the third (the siege ring) is honestly deferred.

One thing the audit did not have to say and this document does: `AuthorityLine.state: REVOKED` is now **observed in a real frame**, so four of that enum's five states are reachable.

End.